

Rookery Bay Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

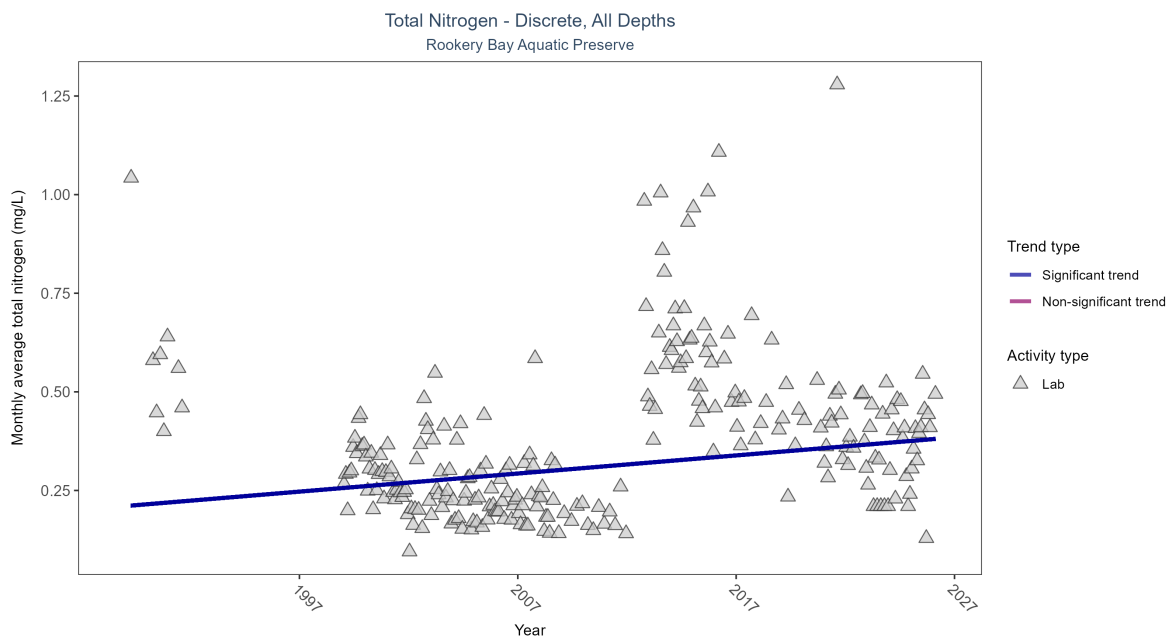


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2653	31	1989 - 2026	0.26	0.16702	0.21032	0.00459	0.0003

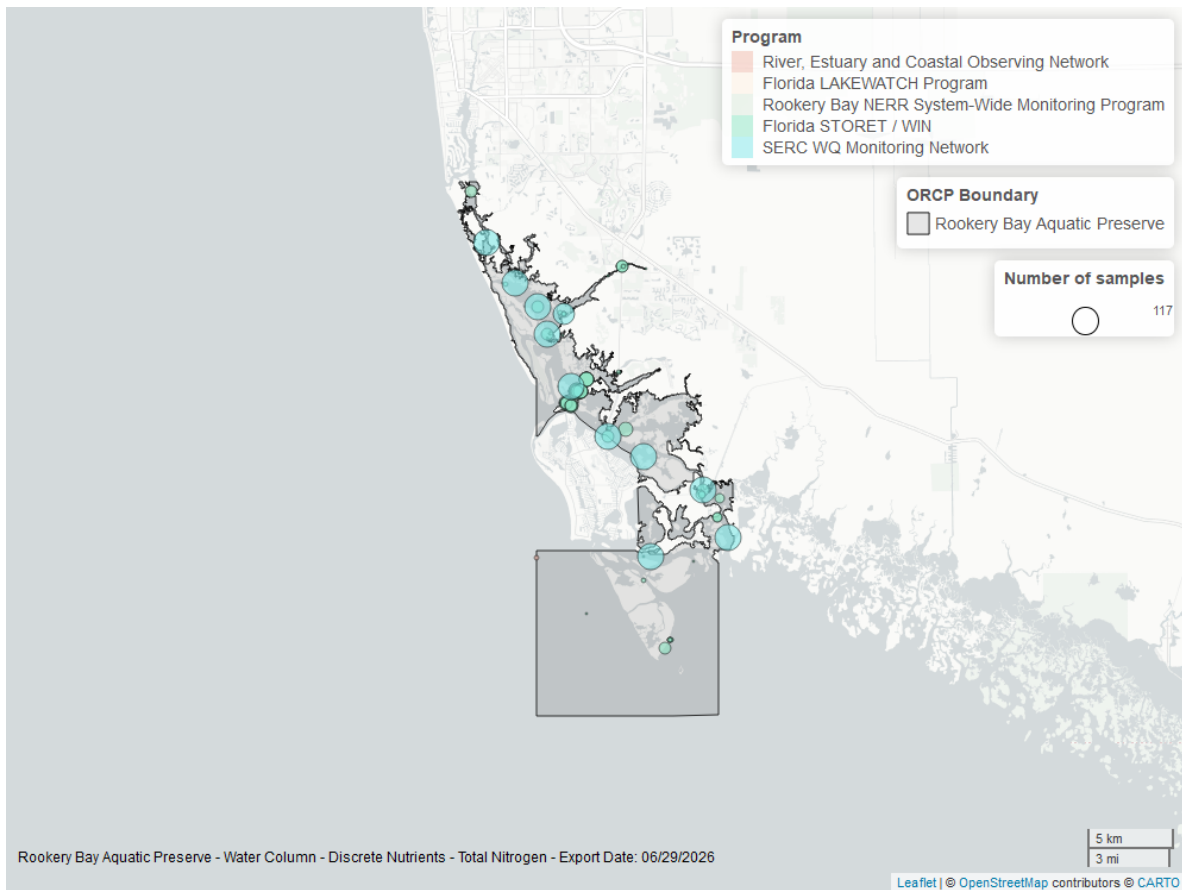


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

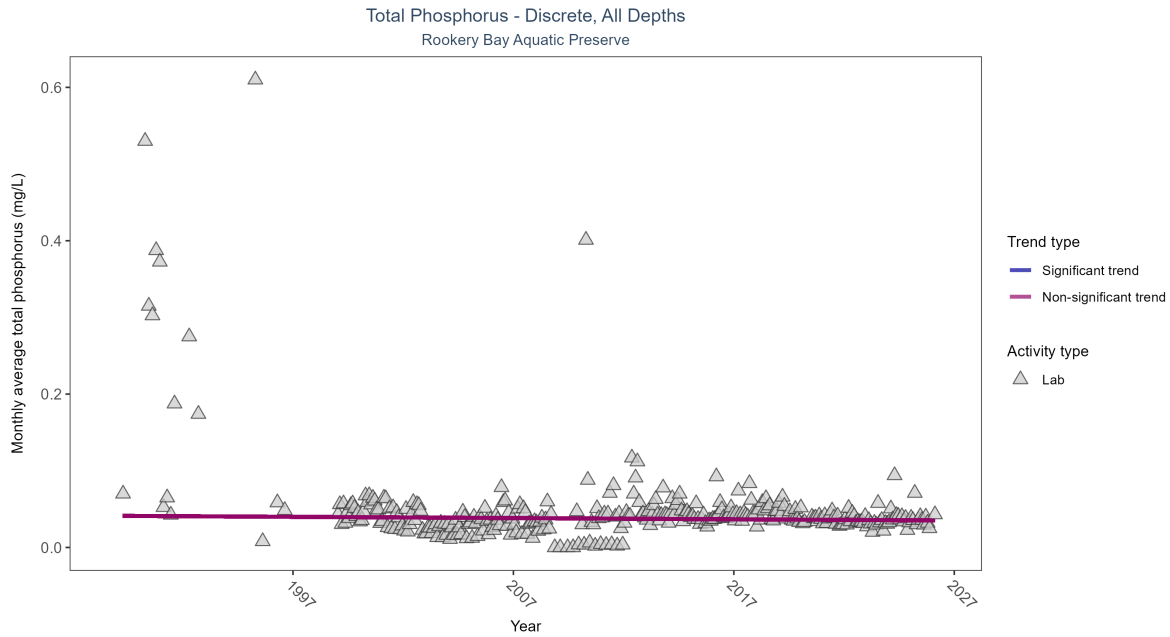


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1989 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2287	34	1989 - 2026	0.034	-0.04792	0.04121	-0.00016	0.1787

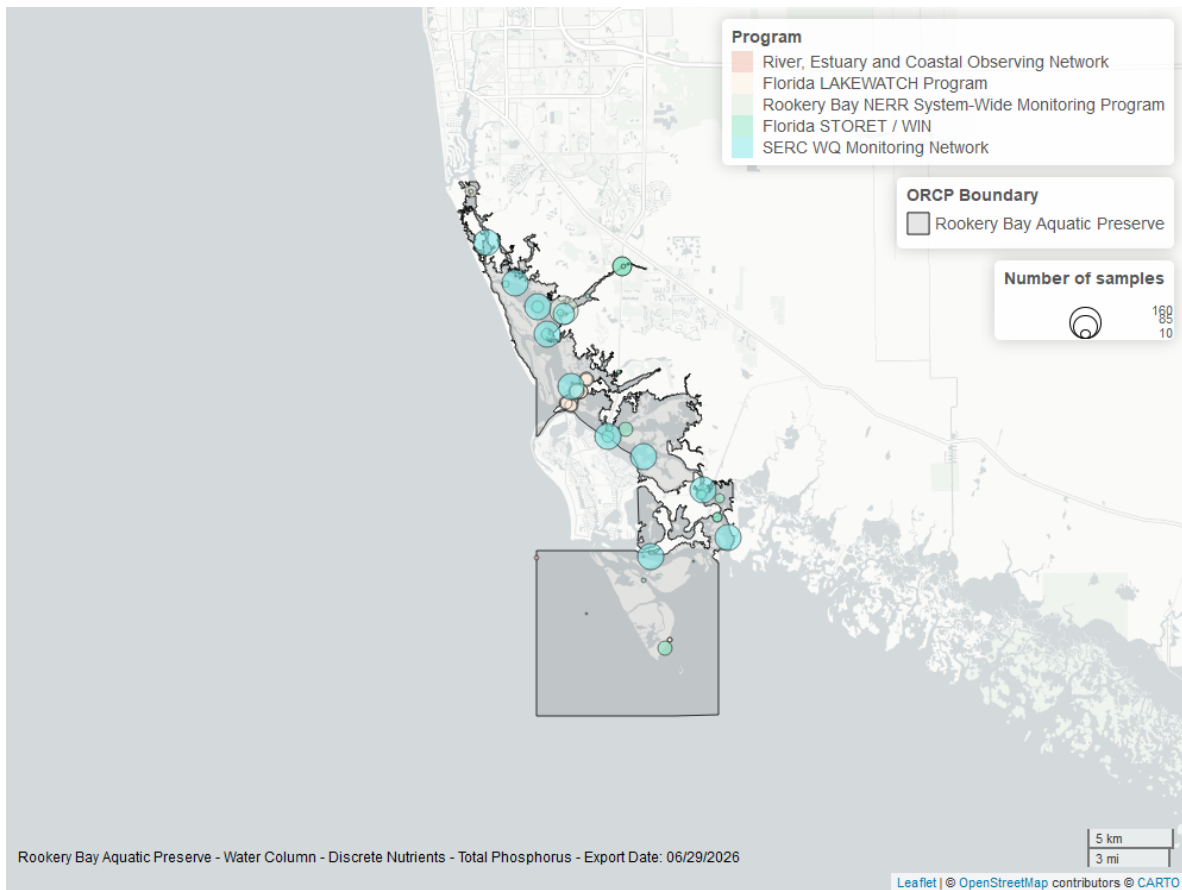


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

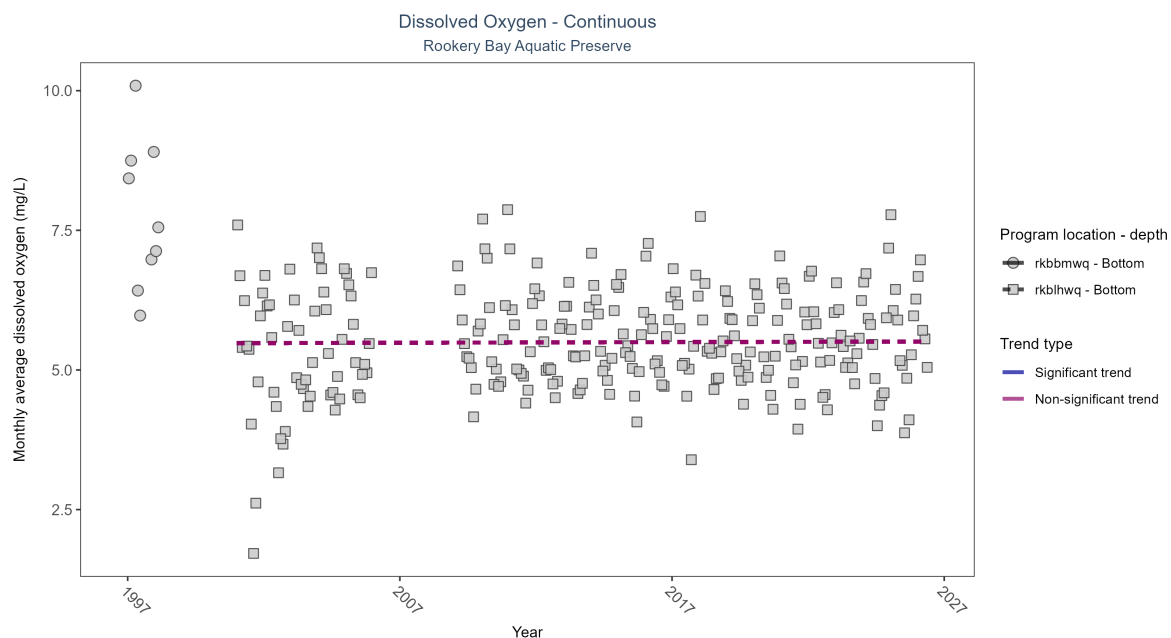


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10441	2	1997 - 1998	7.2	-	-	-	-
rkblhwq	No detectable trend	622816	23	2001 - 2026	5.6	0.01	5.48	0	0.74

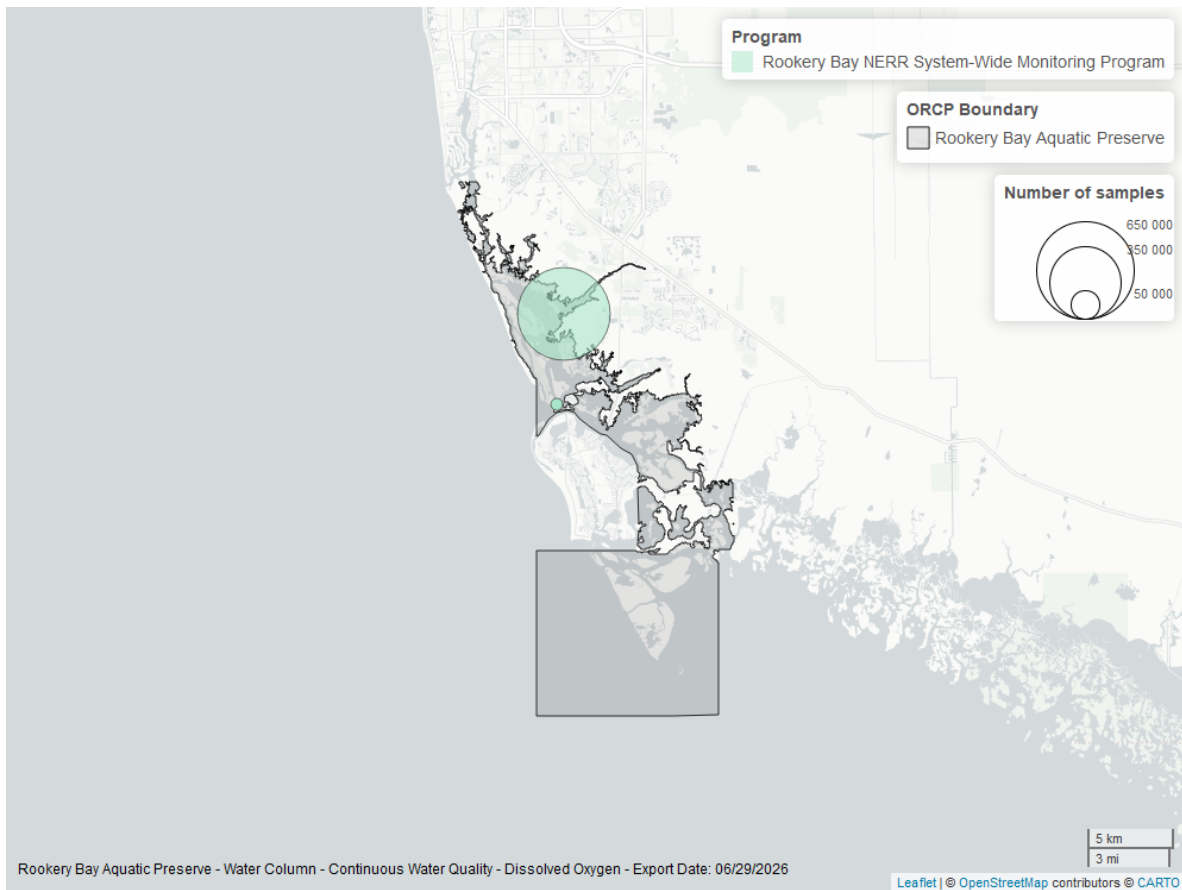


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

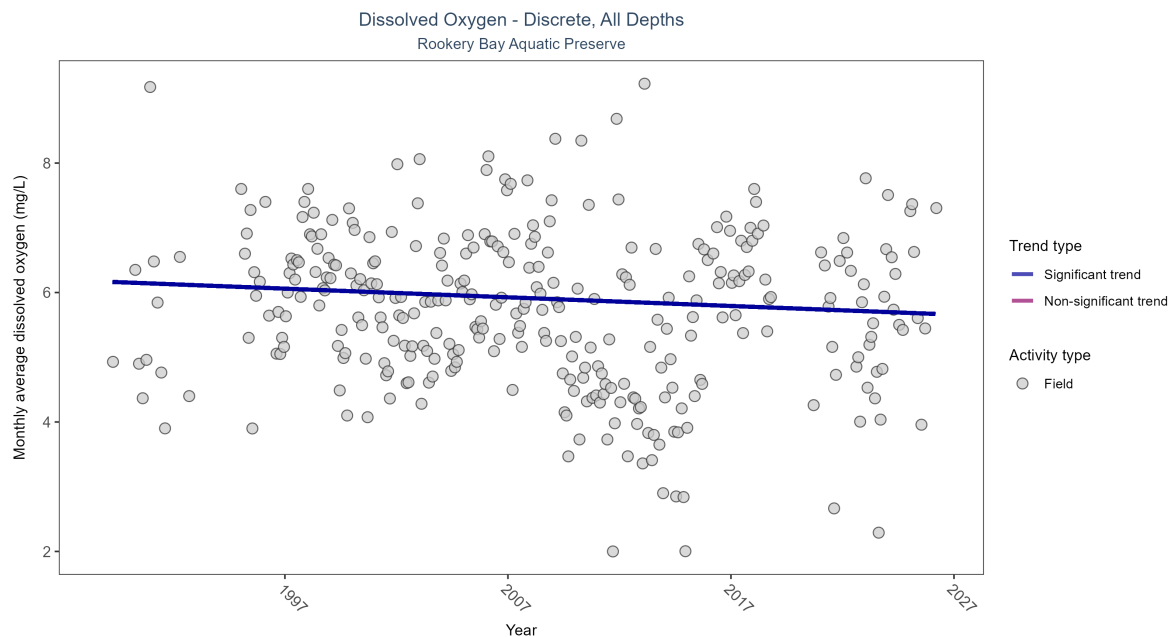


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	3926	35	1989 - 2026	5.9	-0.08699	6.16708	-0.01339	0.0304

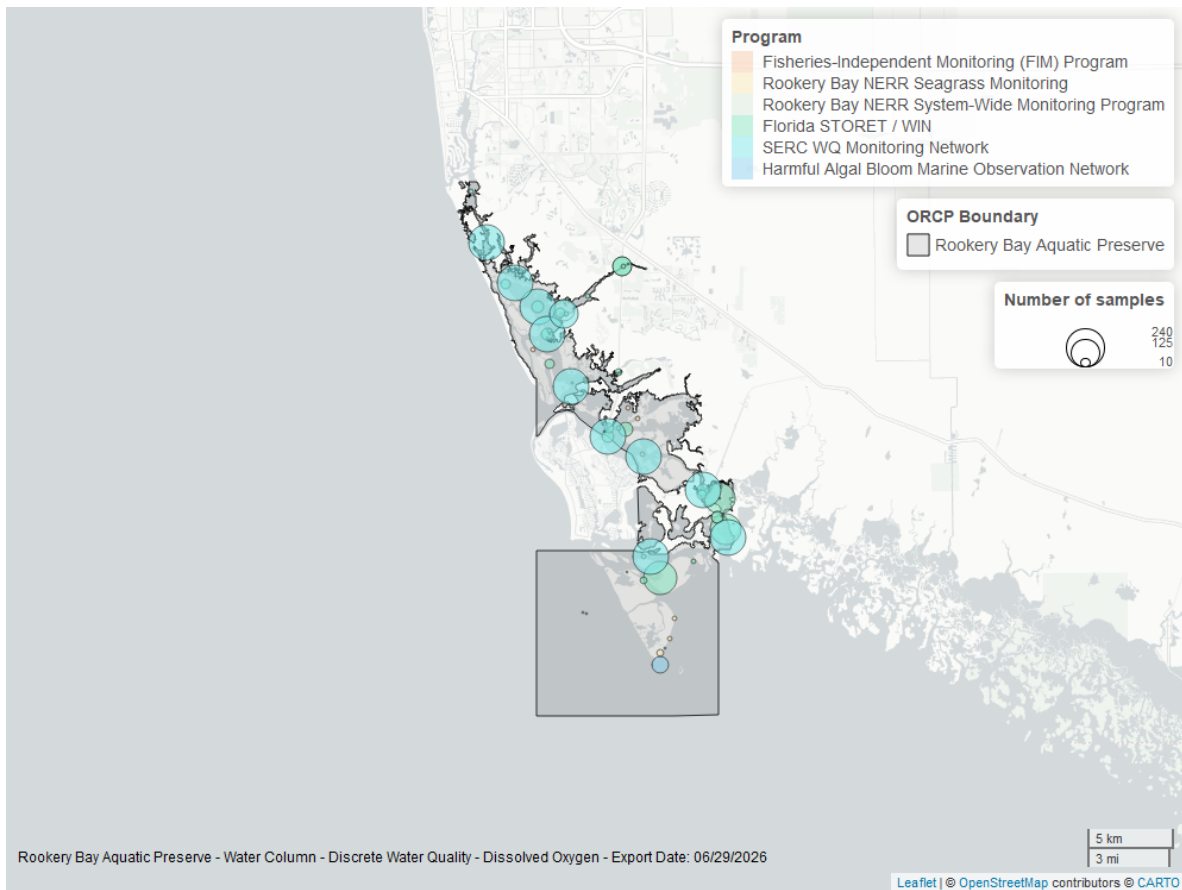


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

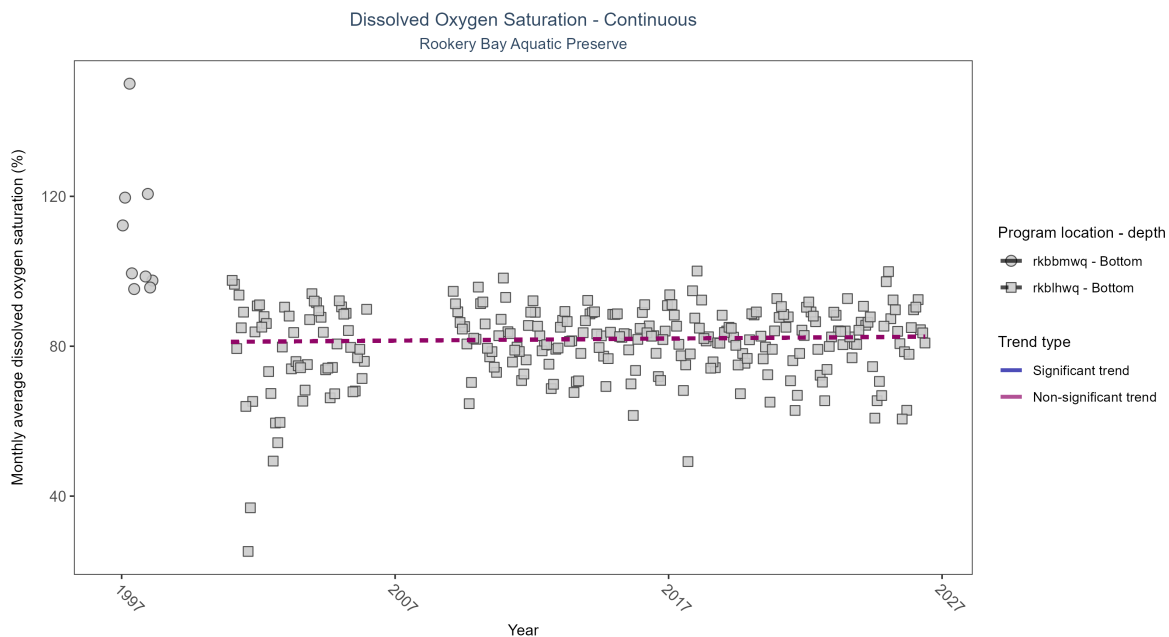


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10441	2	1997 - 1998	102.4	-	-	-	-
rkblhwq	No detectable trend	635134	23	2001 - 2026	82.1	0.05	81.17	0.05	0.22

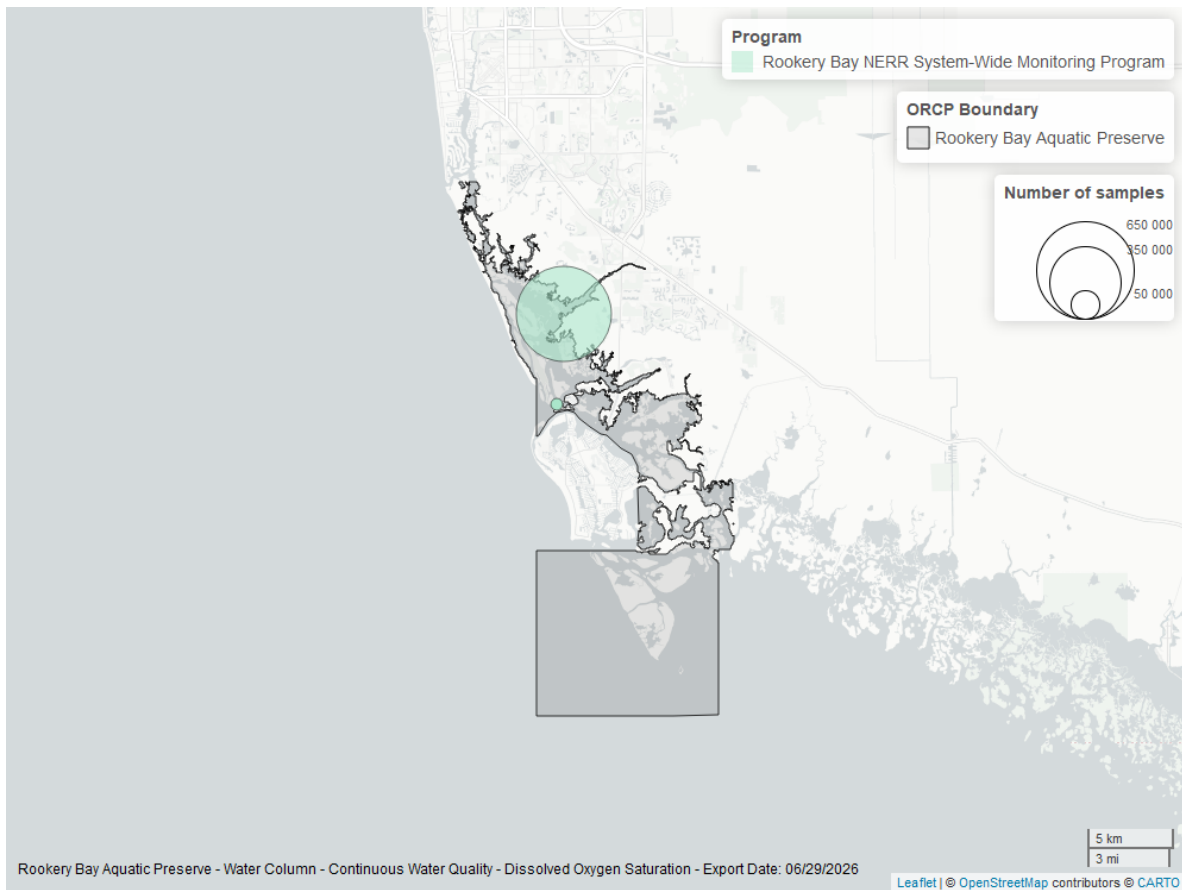


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

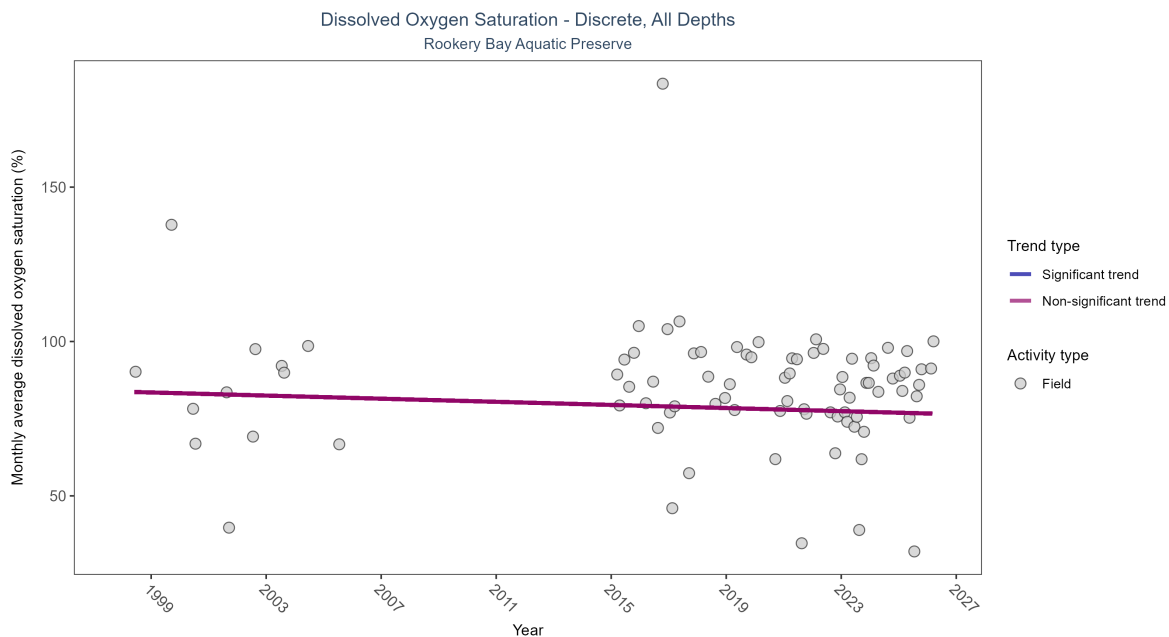


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Dissolved oxygen saturation showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	208	20	1998 - 2026	86.85	-0.05272	83.74348	-0.25217	0.6339

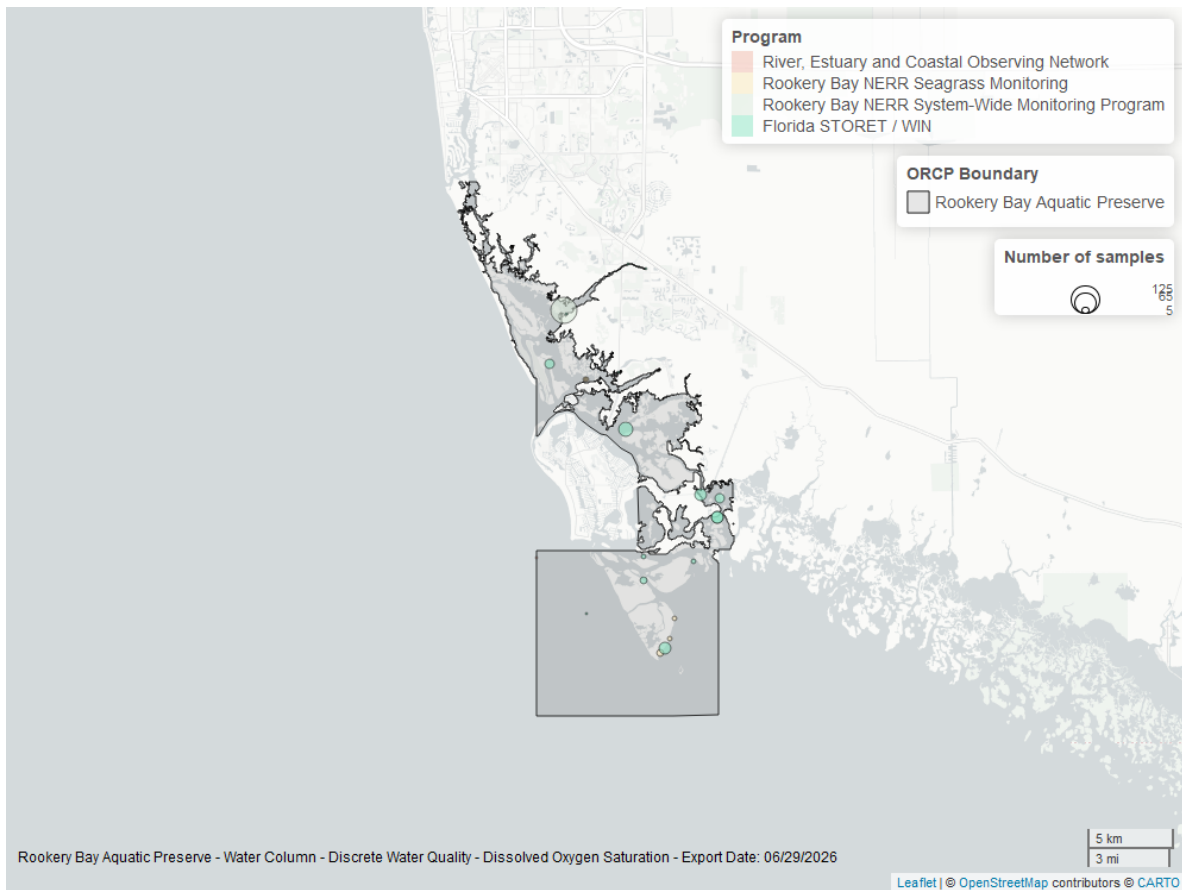


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

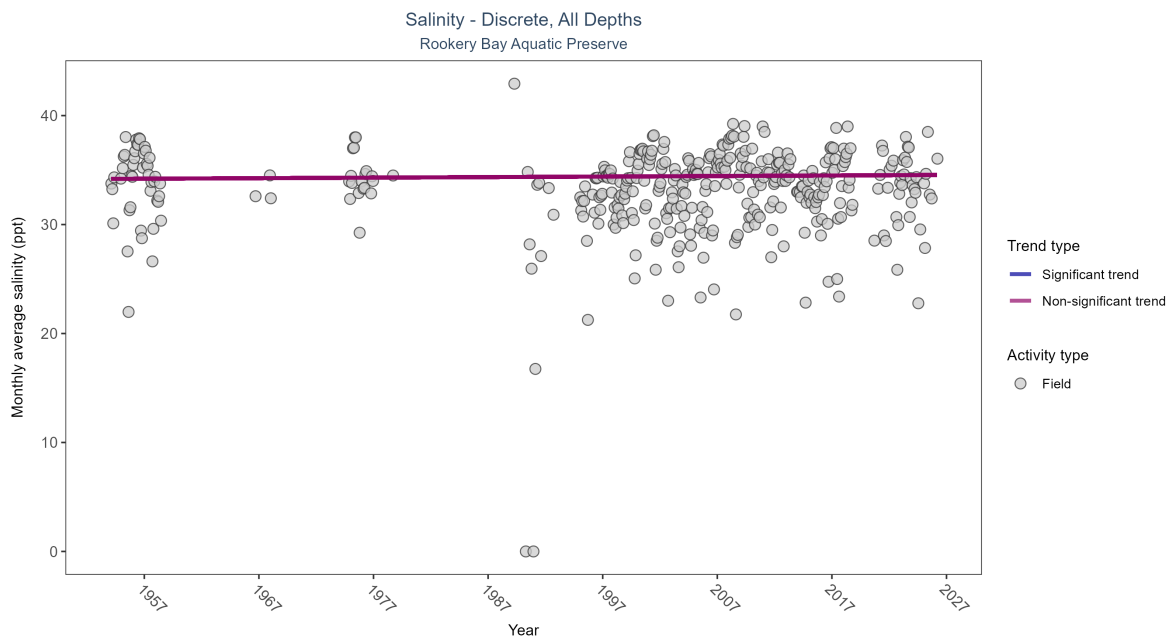


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Salinity showed no detectable trend between 1954 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	4657	47	1954 - 2026	34.24	0.02638	34.19068	0.00494	0.4739

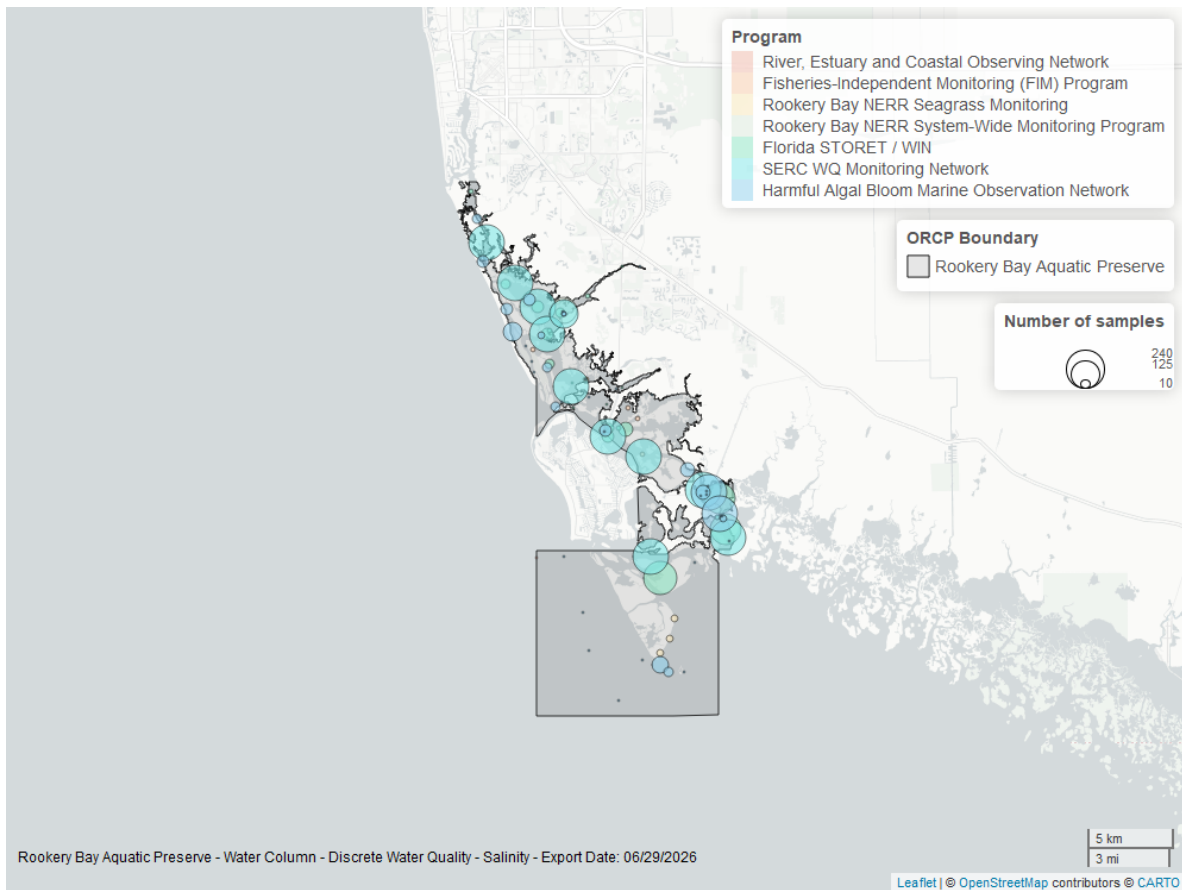


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

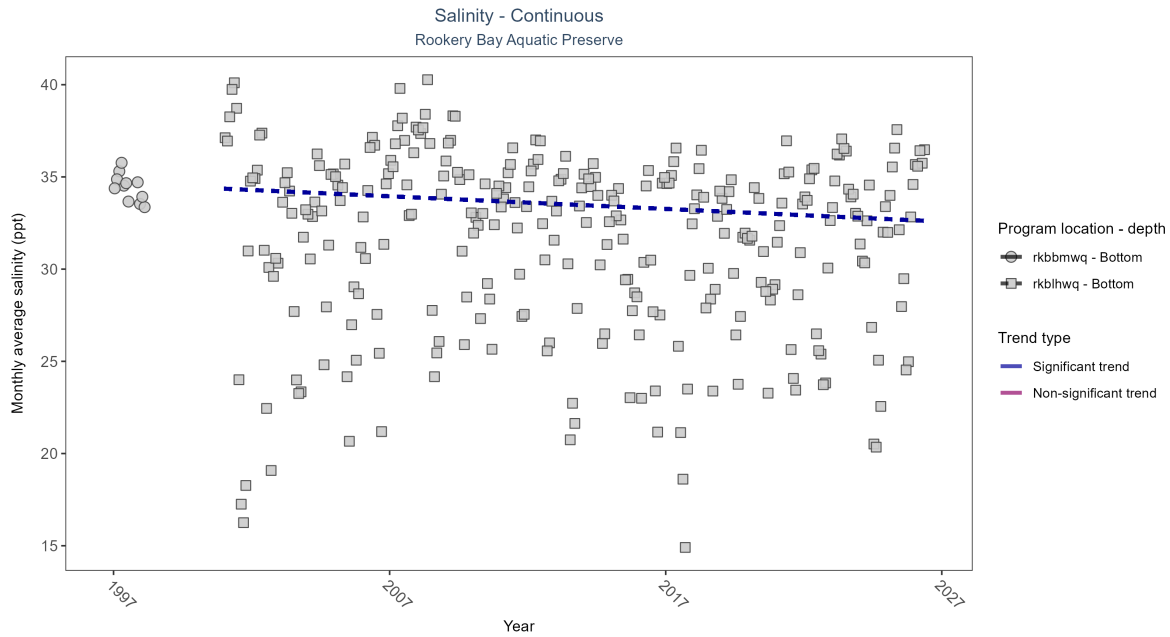


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At one program location, monthly average salinity decreased by 0.07 ppt per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12256	2	1997 - 1998	34.5	-	-	-	-
rkblhwq	Decreasing trend	709967	26	2001 - 2026	33.2	-0.12	34.36	-0.07	0

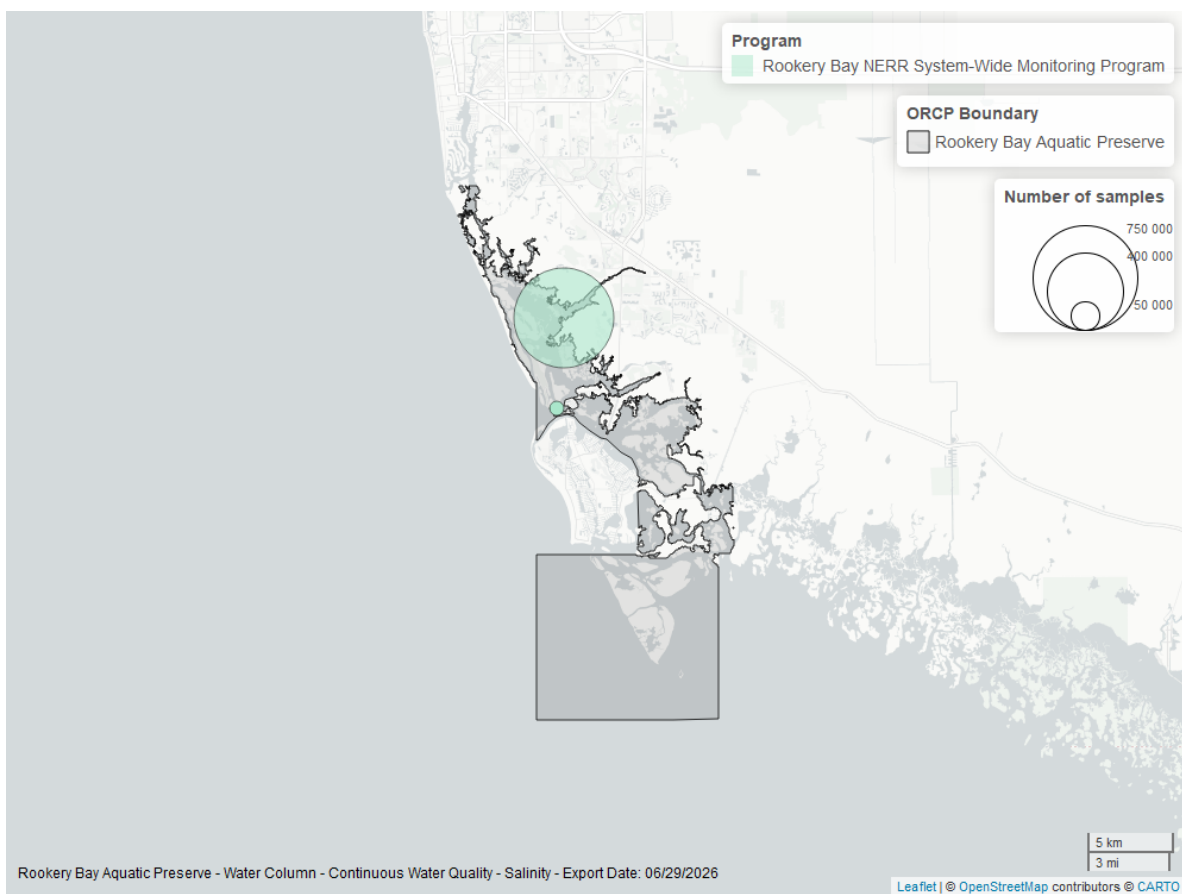


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

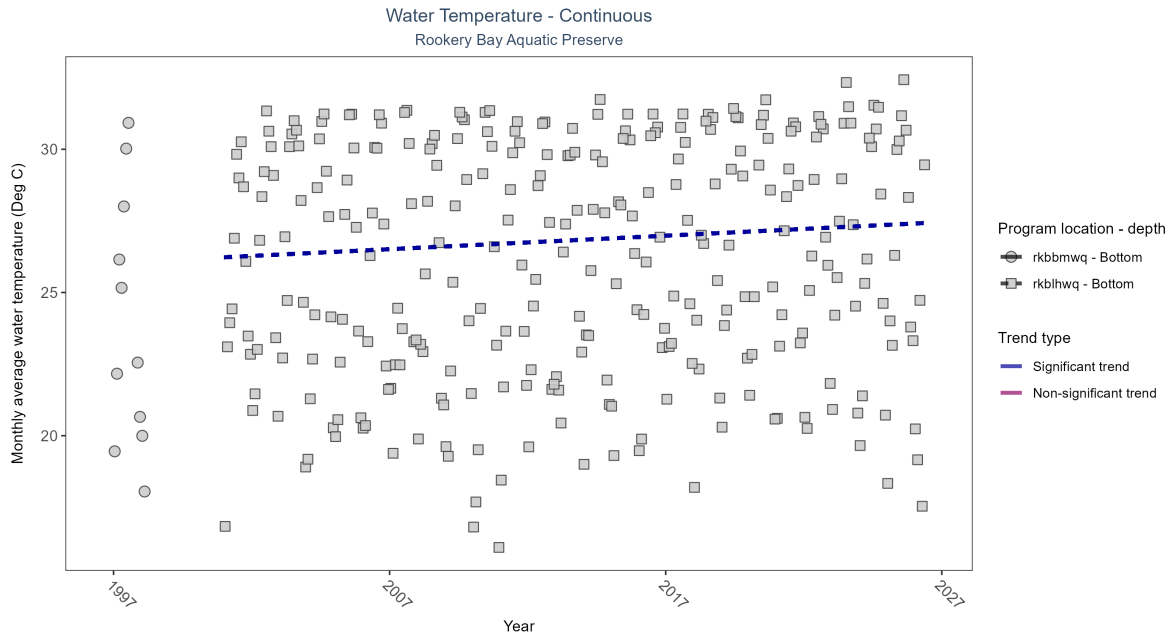


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At one program location, monthly average water temperature increased by 0.05Deg C per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12610	2	1997 - 1998	23.8	-	-	-	-
rkblhwq	Increasing trend	741119	26	2001 - 2026	26.8	0.23	26.23	0.05	0

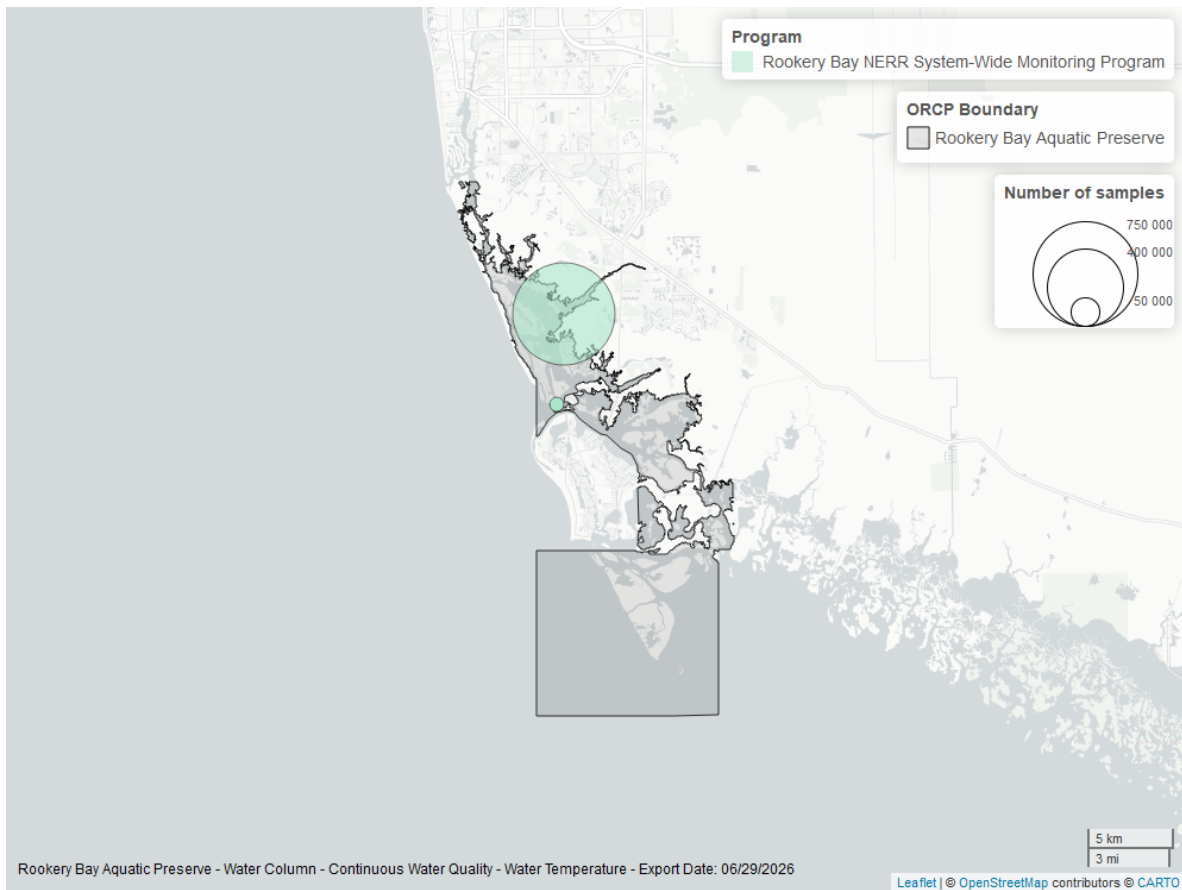


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

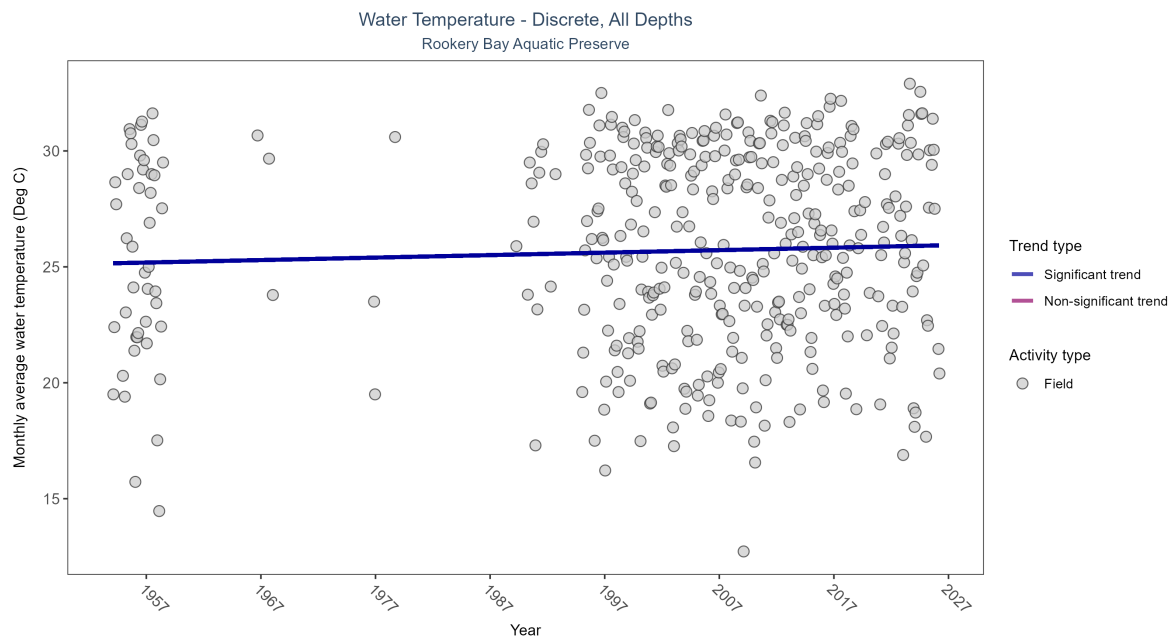


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Monthly average water temperature increased by 0.01Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	4592	46	1954 - 2026	26.2	0.08211	25.15326	0.01069	0.0148

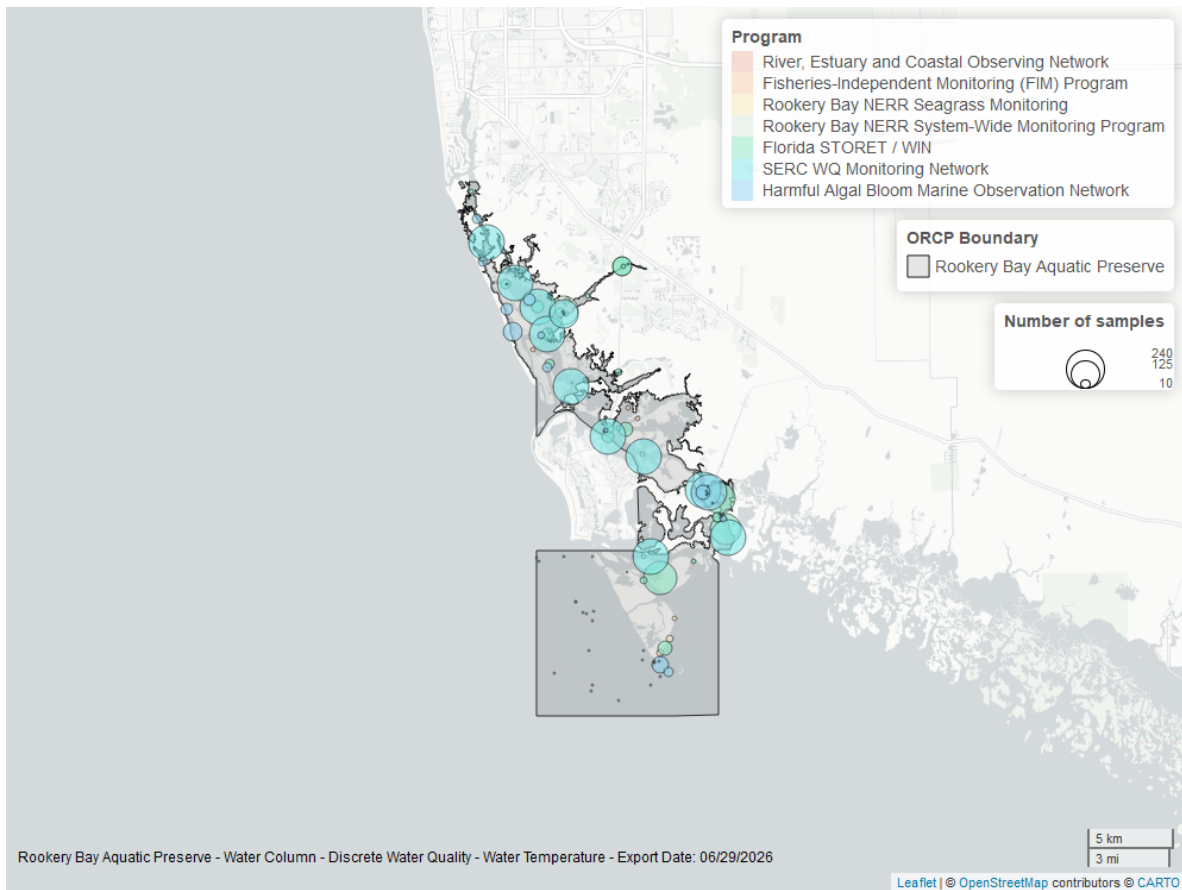


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

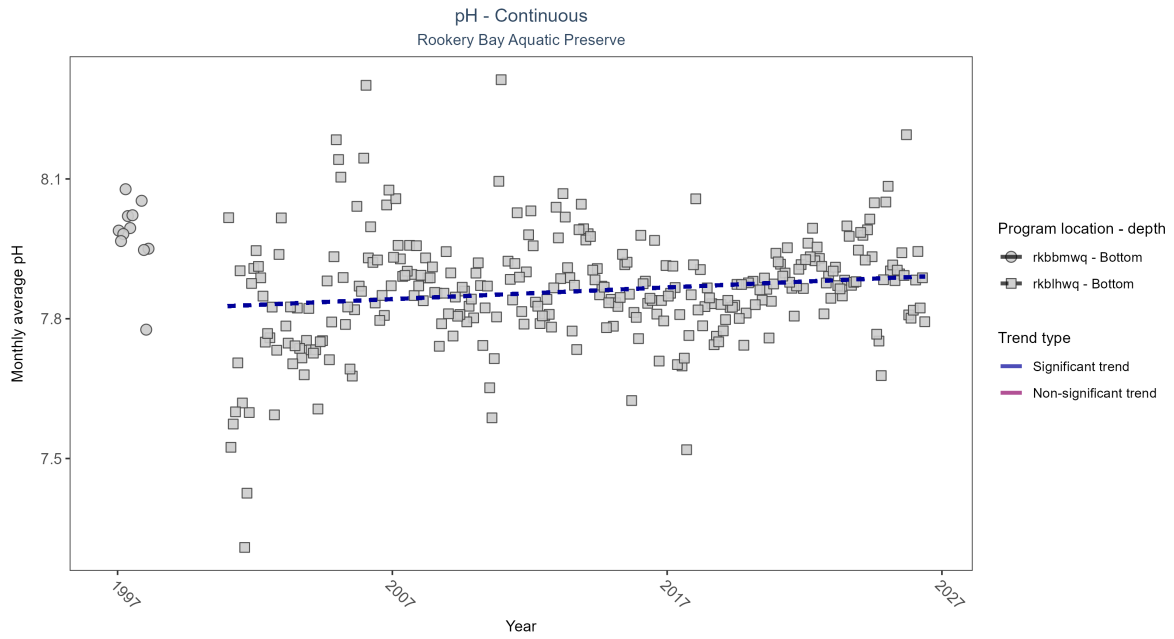


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At one program location, monthly average pH increased by less than 0.01 pH units per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12610	2	1997 - 1998	8.0	-	-	-	-
rkblhwq	Increasing trend	680804	26	2001 - 2026	7.9	0.14	7.83	0	0

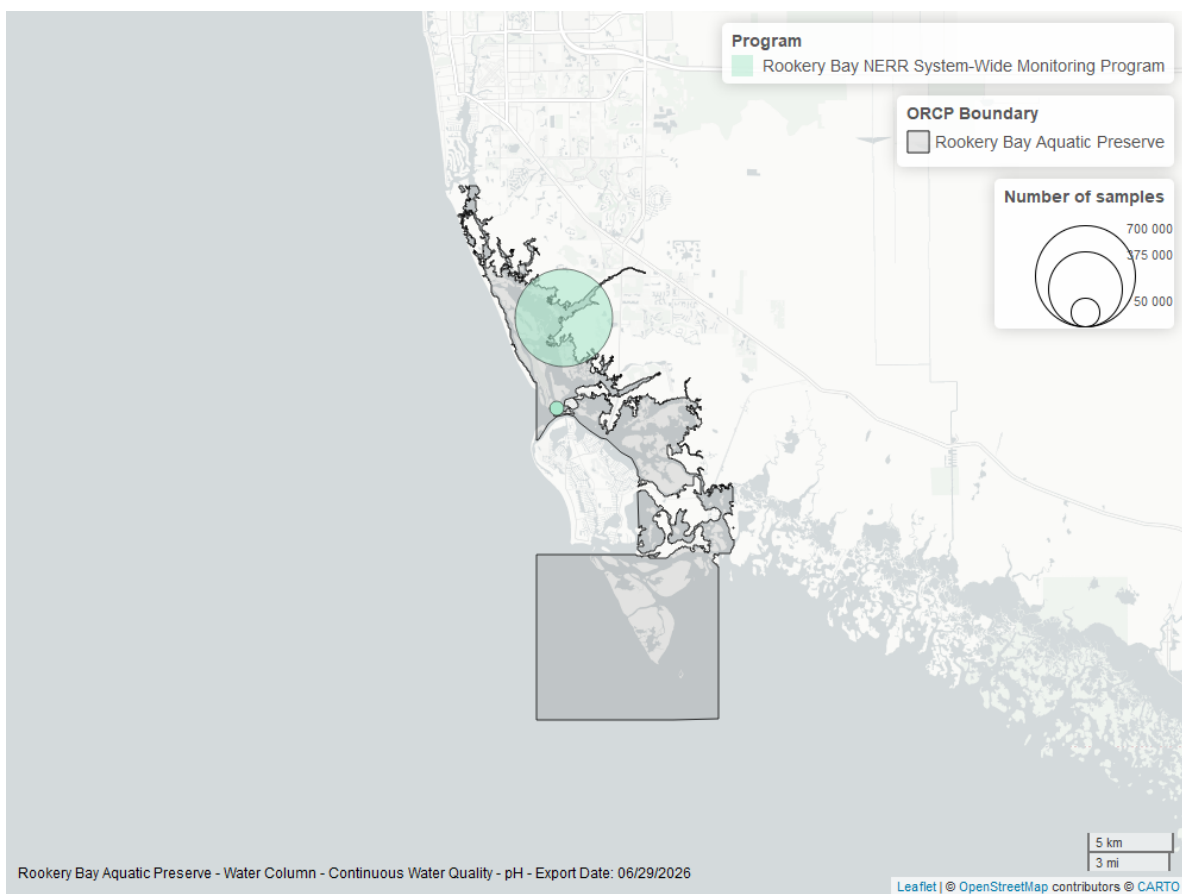


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

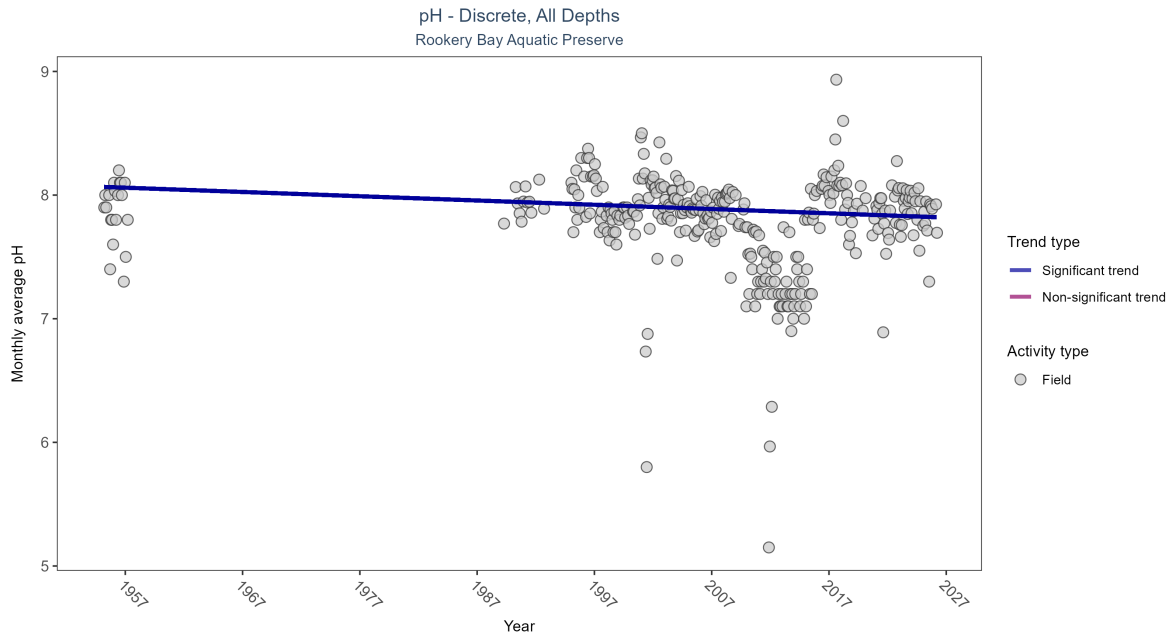


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	1898	39	1955 - 2026	7.9	-0.13807	8.06668	-0.00345	0.0003

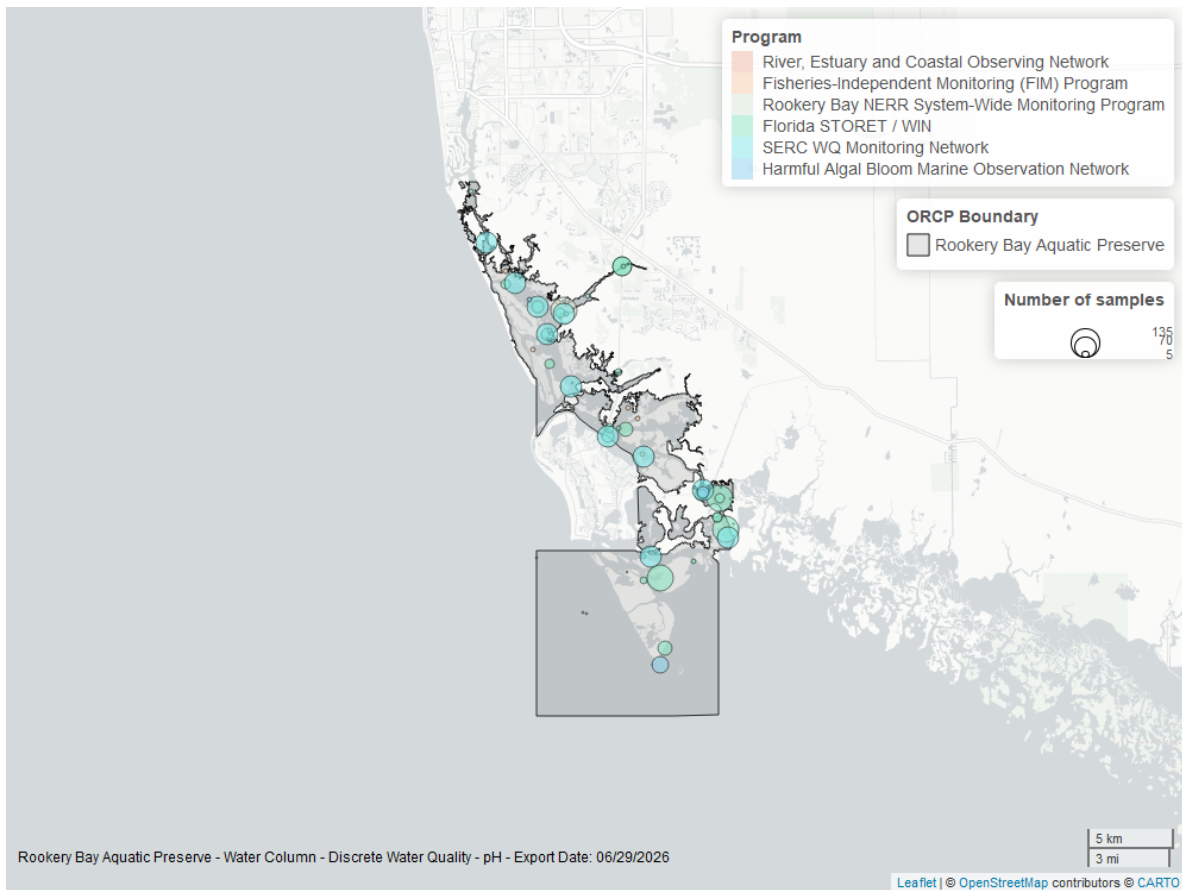


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

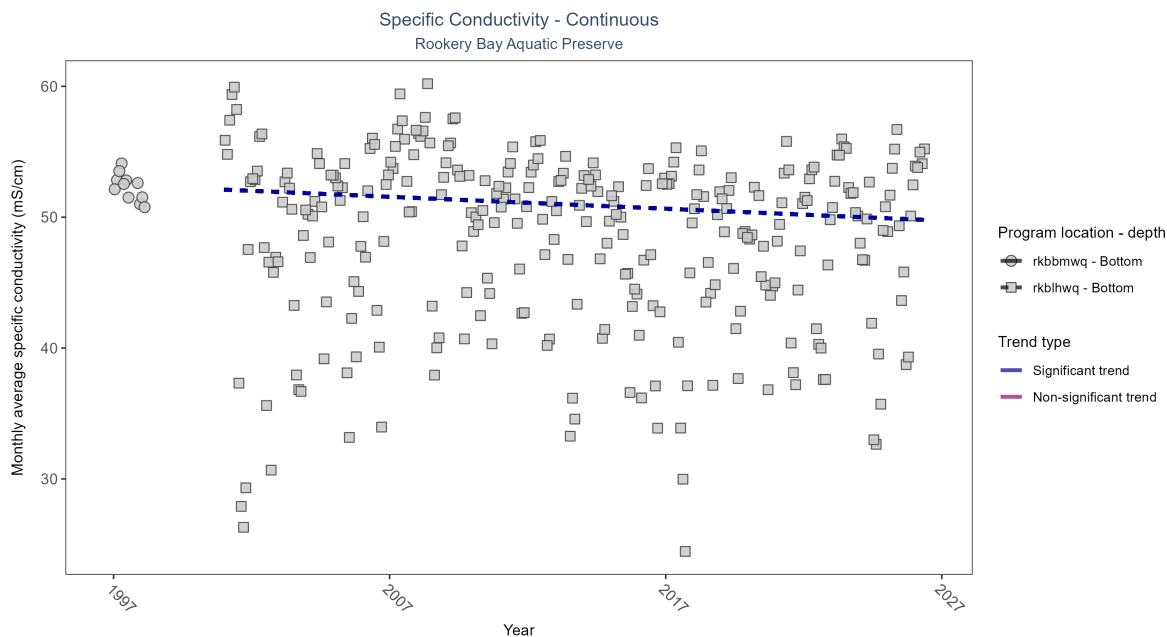


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At one program location, monthly average specific conductivity decreased by 0.09 mS/cm per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12256	2	1997 - 1998	52.41	-	-	-	-
rkblhwq	Decreasing trend	710006	26	2001 - 2026	50.68	-0.11	52.11	-0.09	0.01

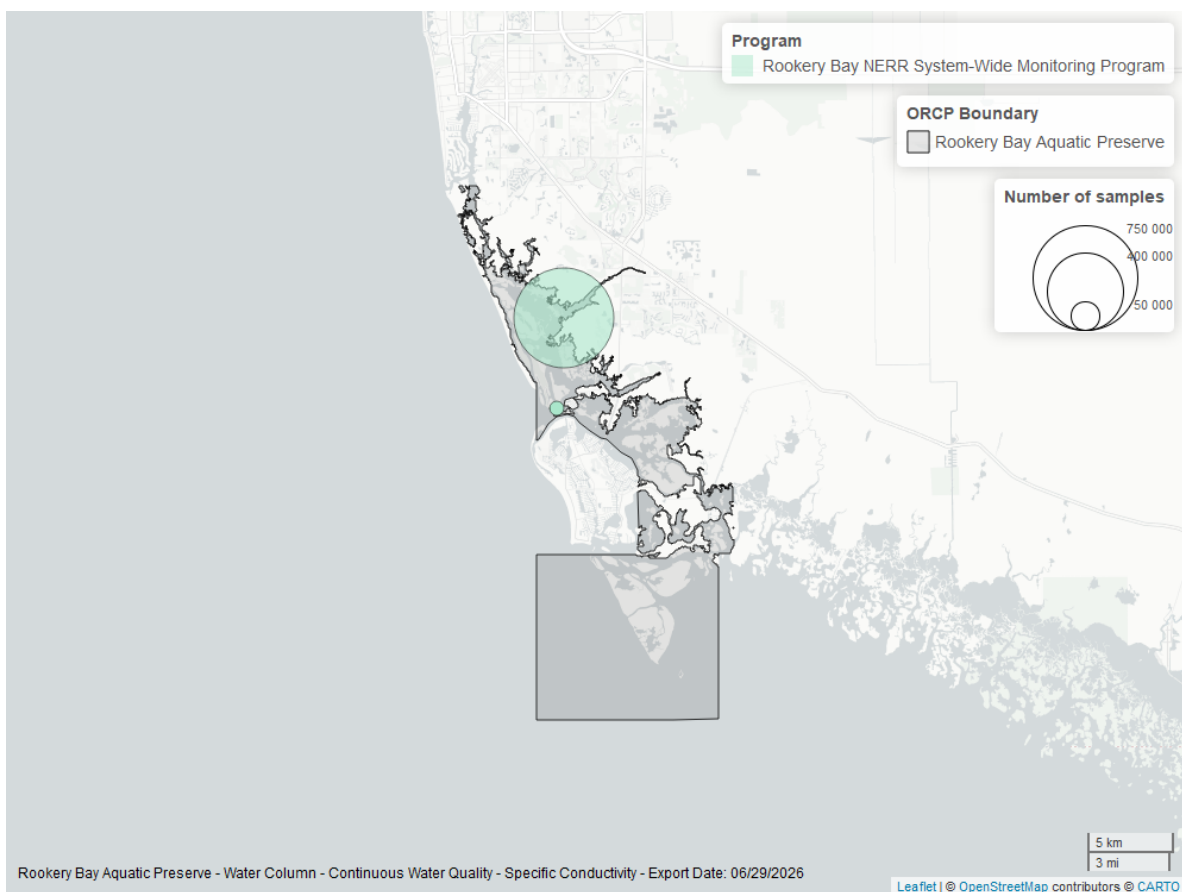


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

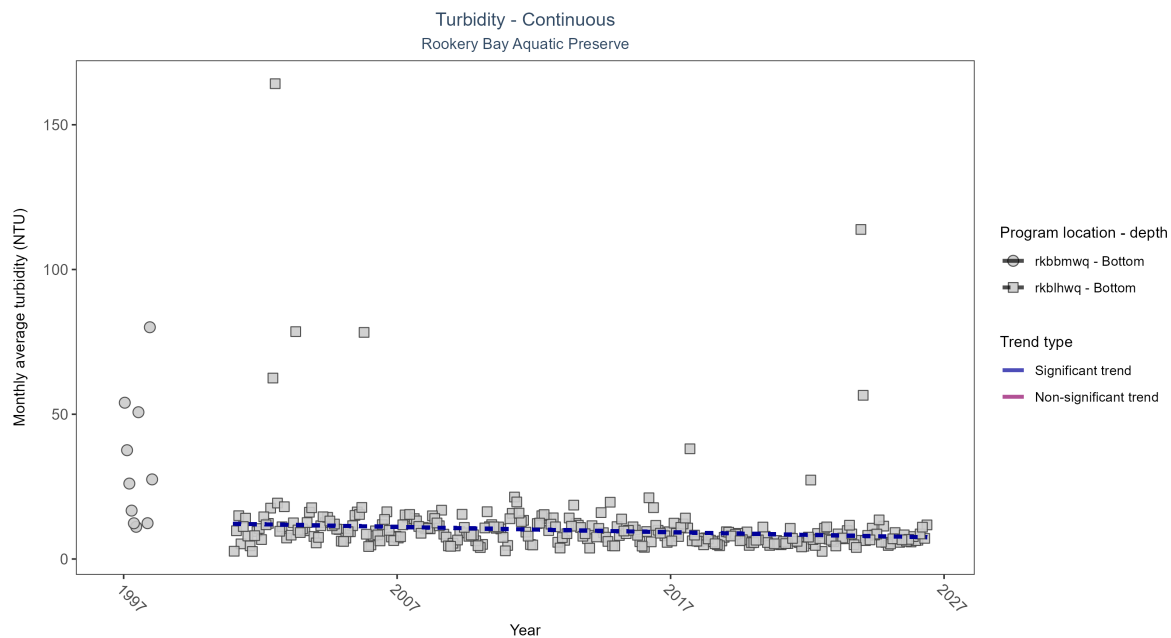


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At one program location, monthly average turbidity decreased by 0.18 NTU per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10654	2	1997 - 1998	11	-	-	-	-
rkblhwq	Decreasing trend	657127	26	2001 - 2026	8	-0.27	12.18	-0.18	0

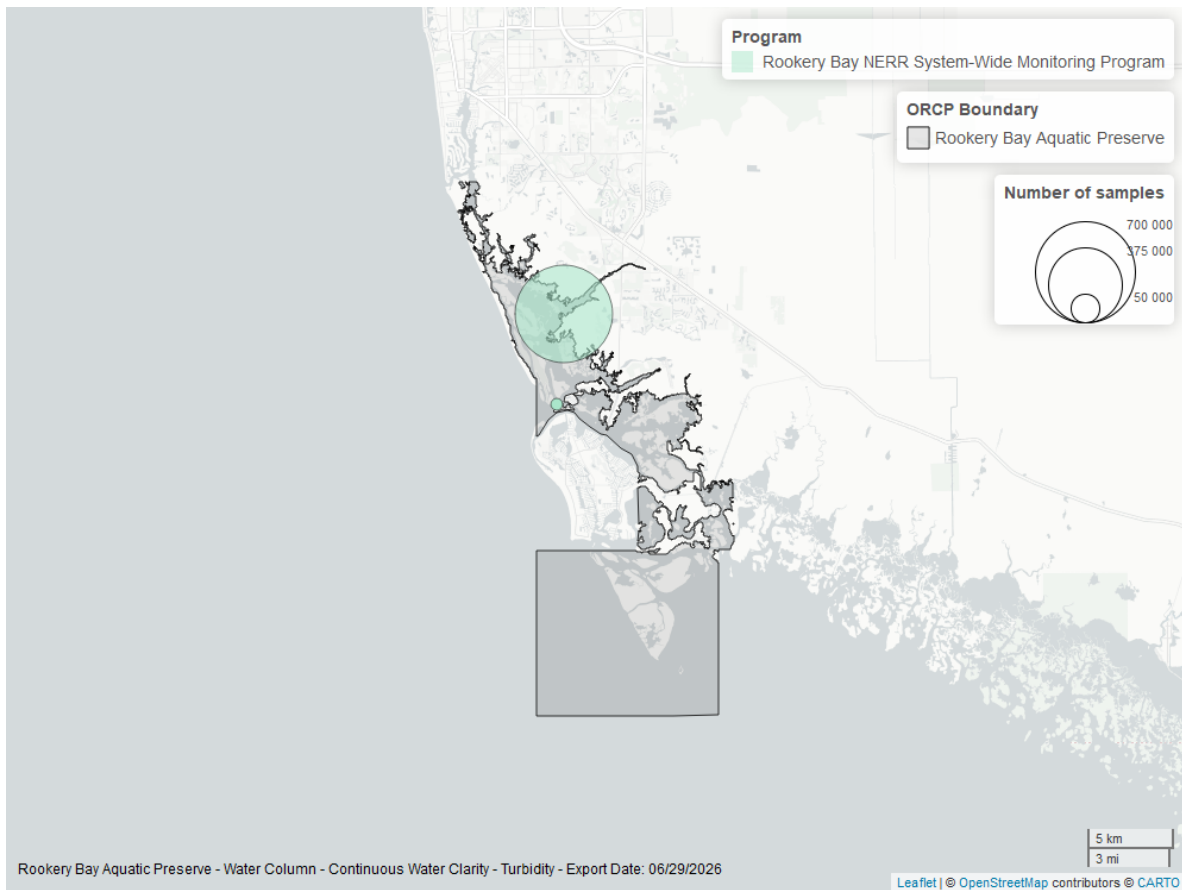


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

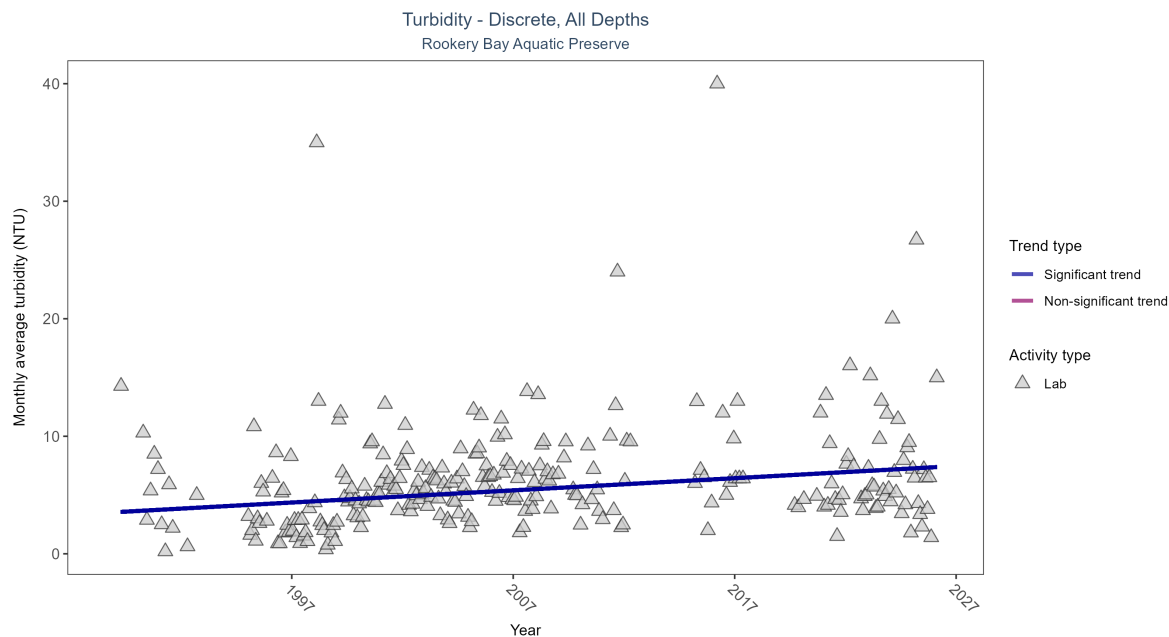


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Monthly average turbidity increased by 0.1 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1892	33	1989 - 2026	4.9925	0.21521	3.5342	0.10353	0

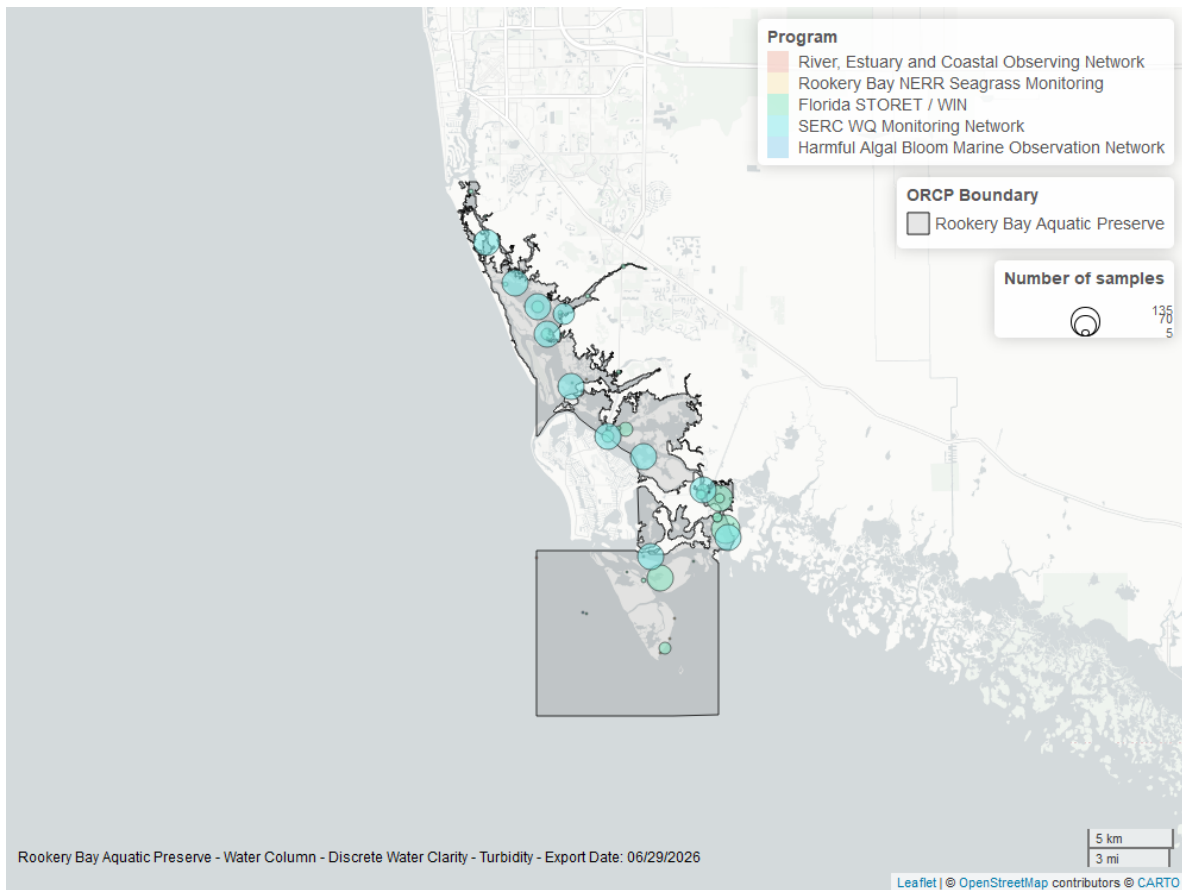


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

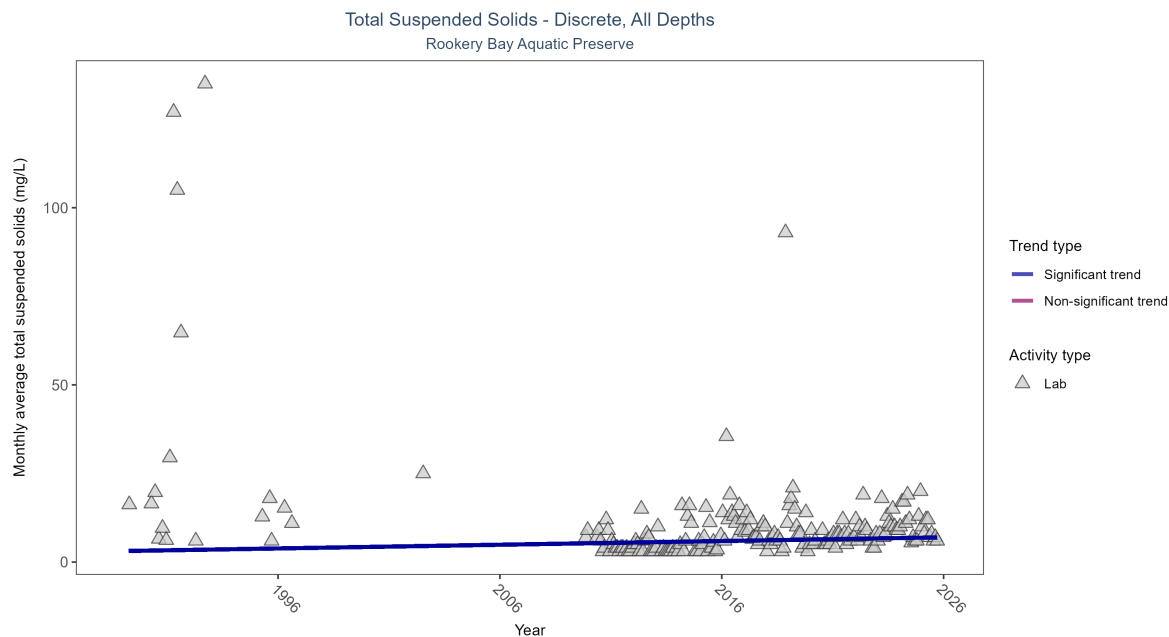


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Monthly average total suspended solids increased by 0.1 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	265	24	1989 - 2025	8	0.12857	3.1125	0.10455	0.0345

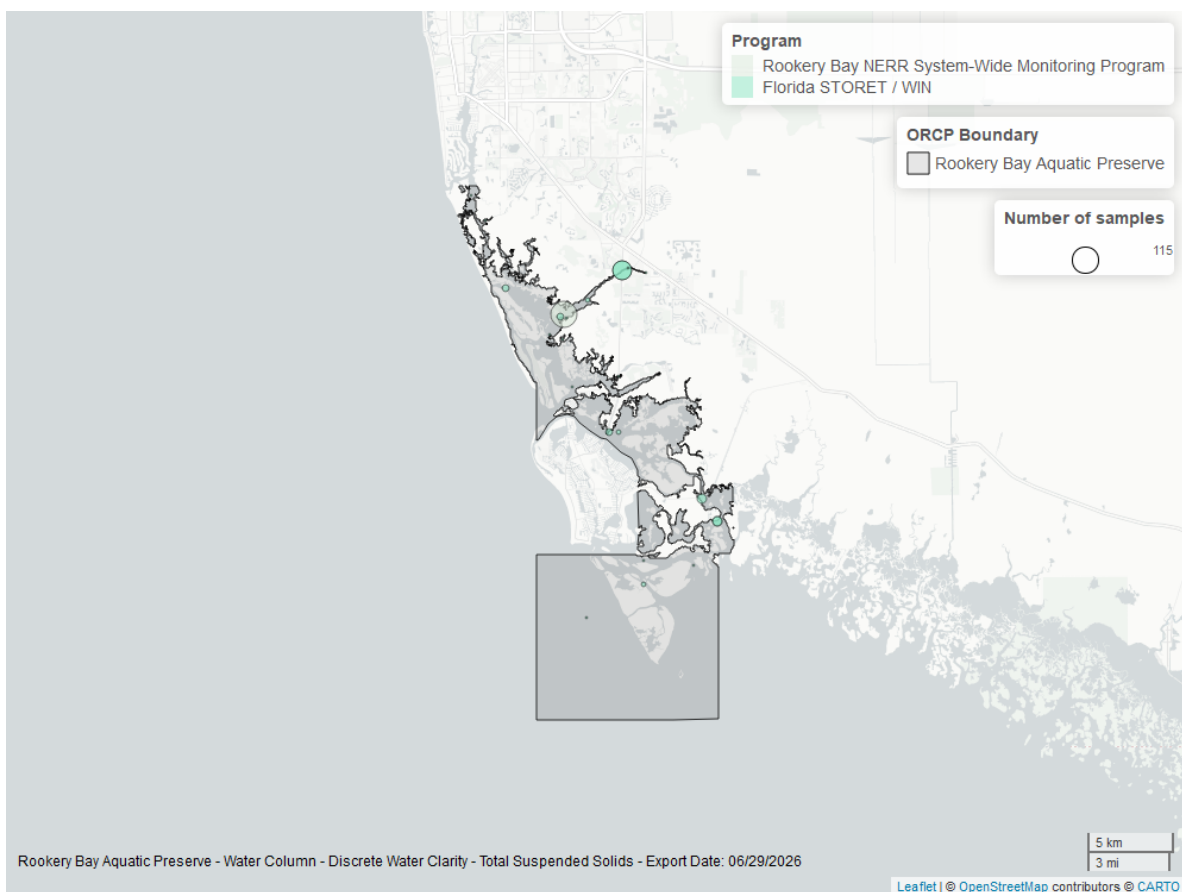


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

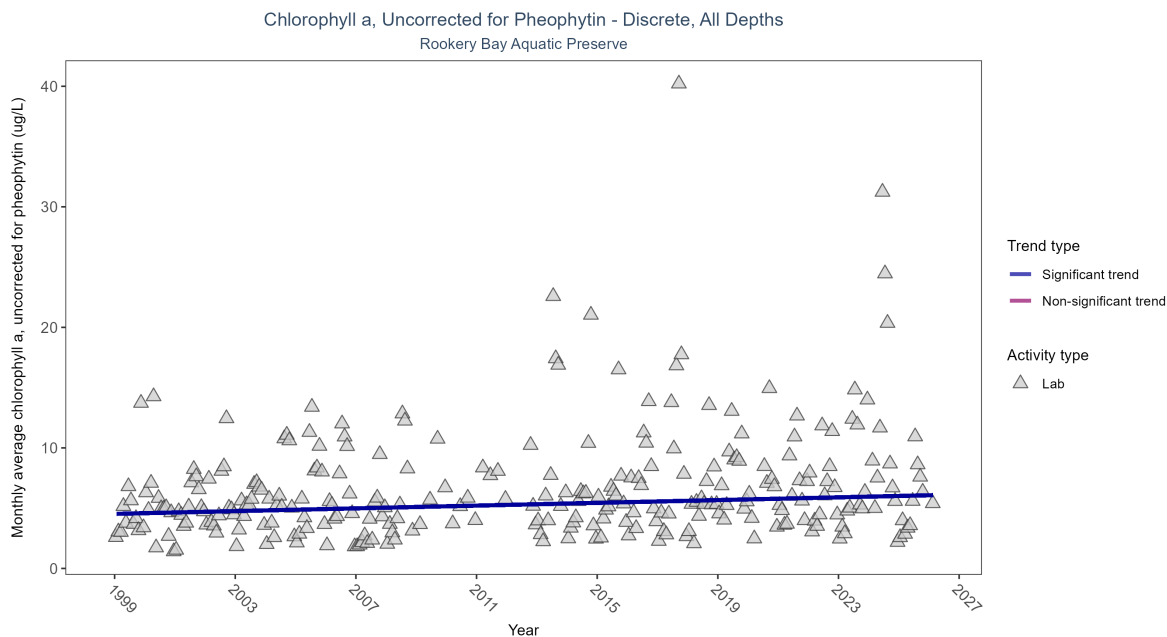


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.06 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3329	28	1999 - 2026	5	0.17399	4.52693	0.05711	0.0001

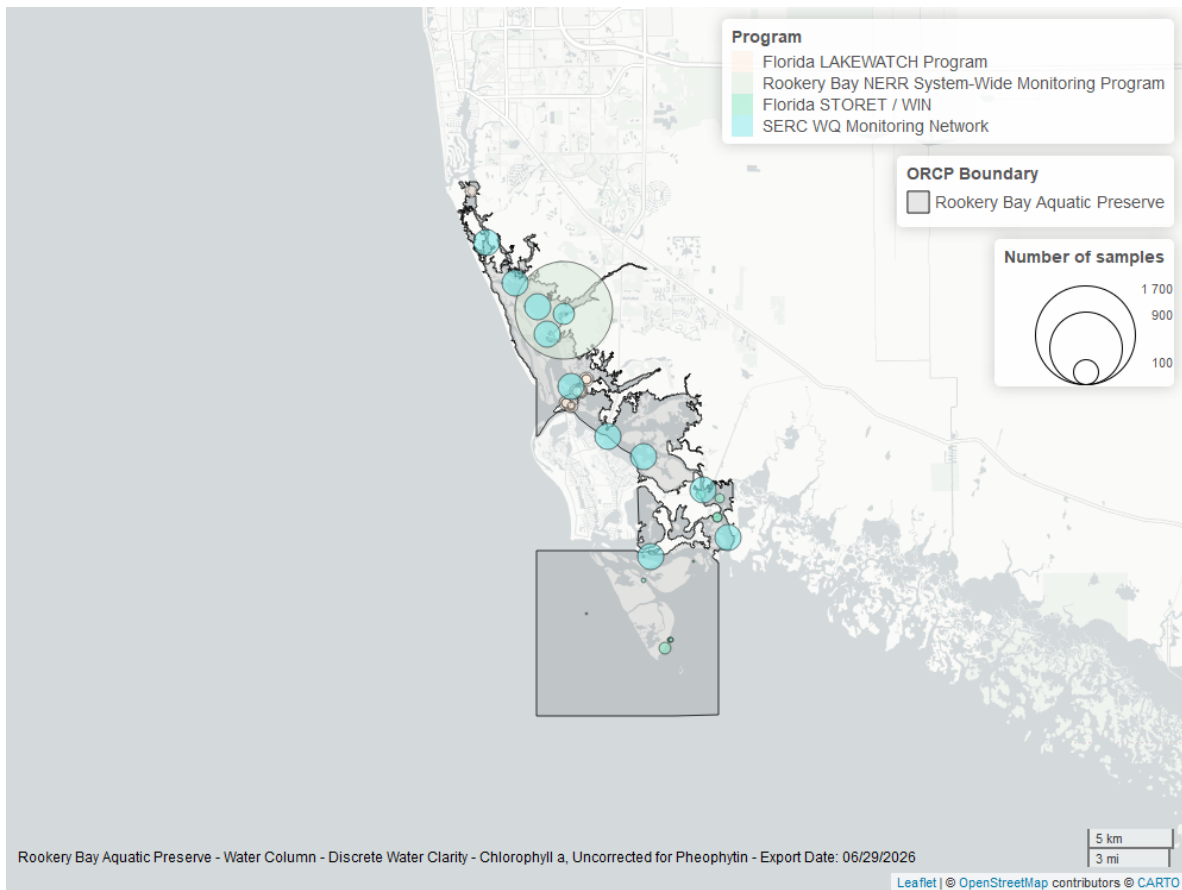


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

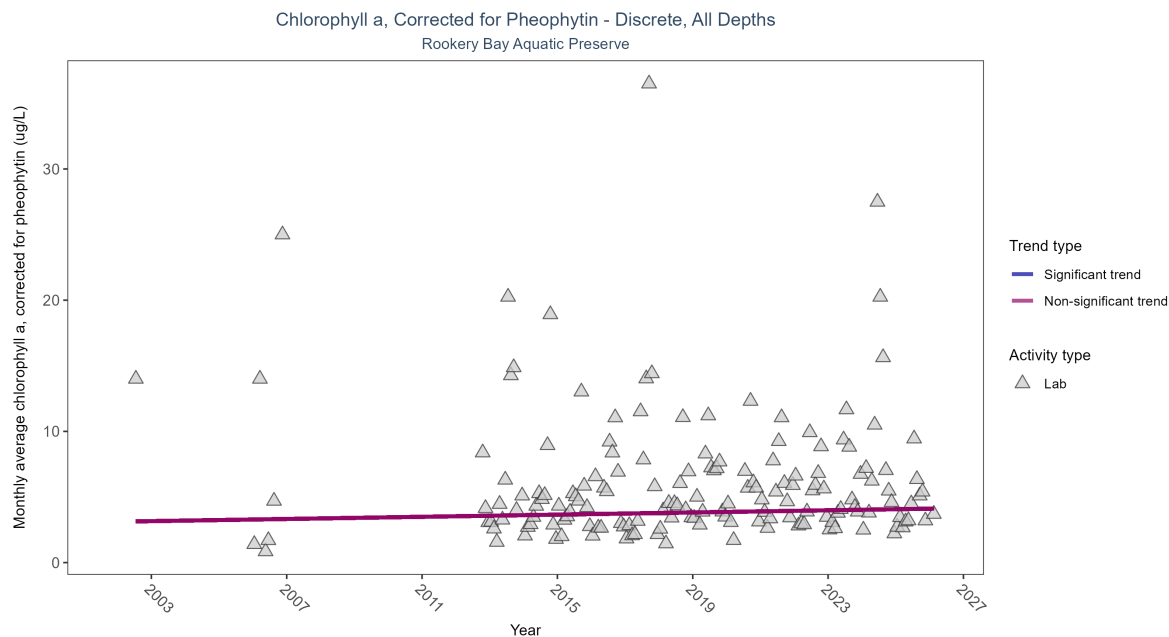


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1762	17	2002 - 2026	4.3	0.07653	3.11785	0.0413	0.1949

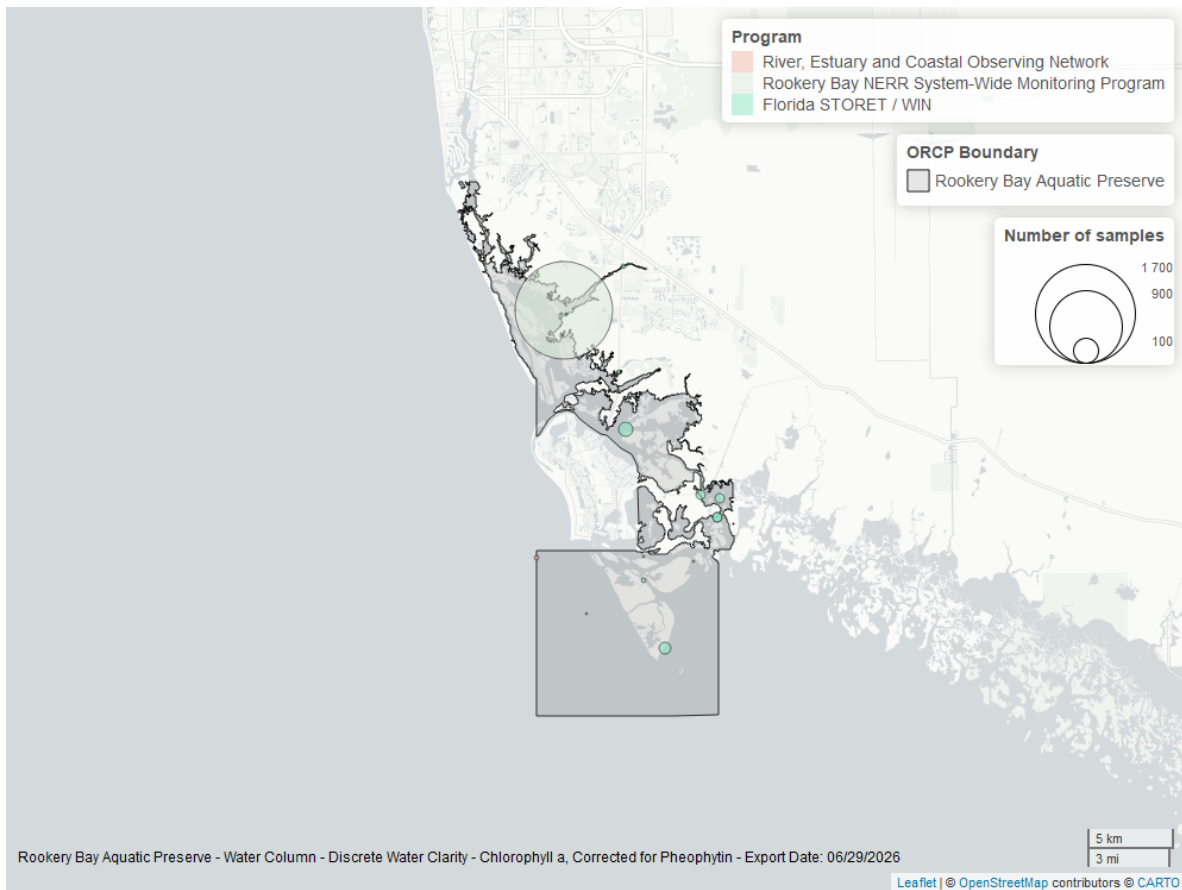


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

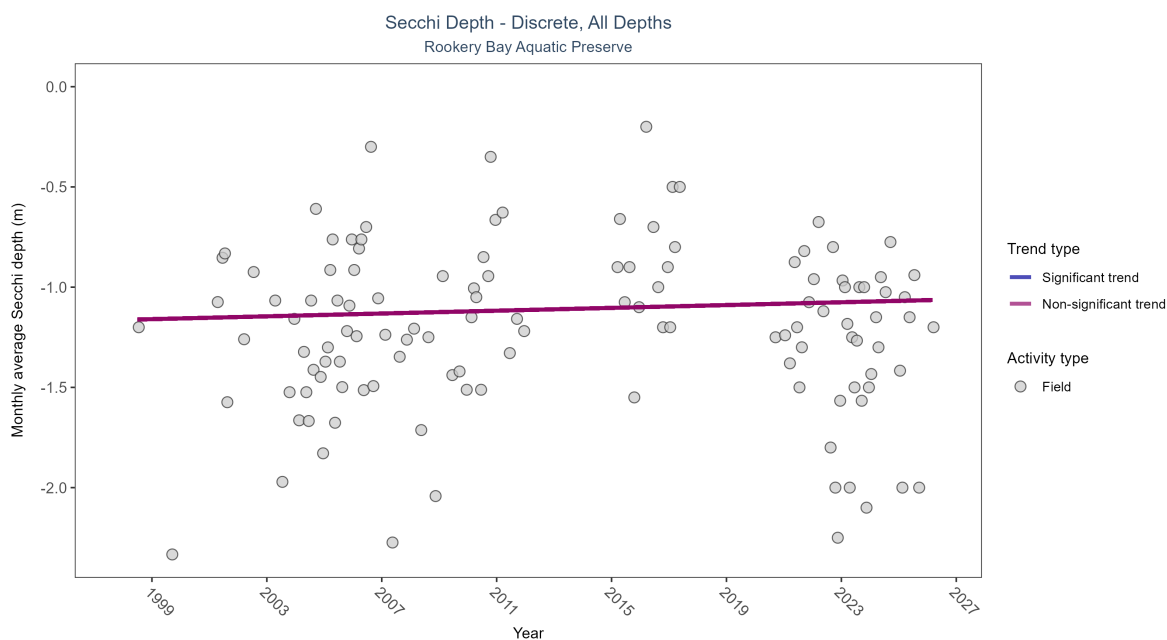


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Secchi depth showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	442	23	1998 - 2026	-1.2	0.03973	-1.16309	0.00351	0.5308

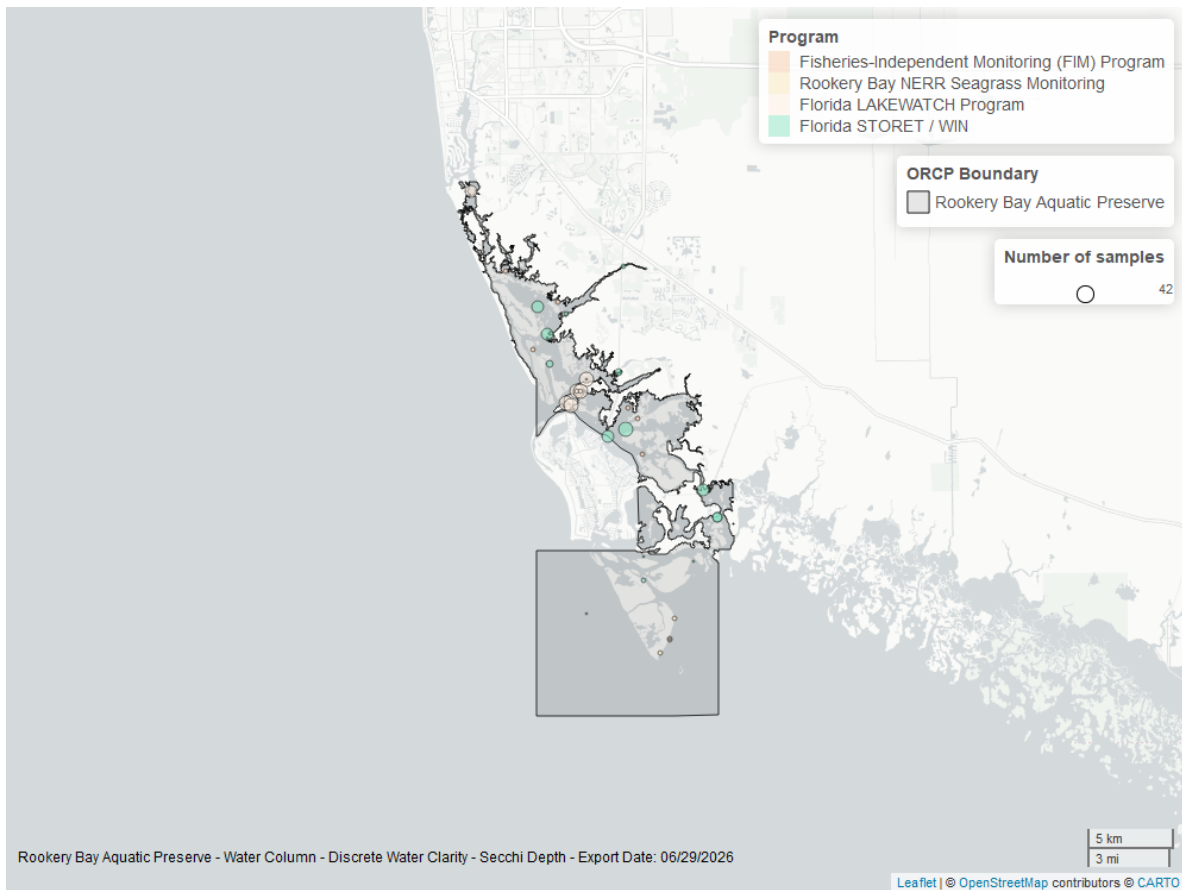


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

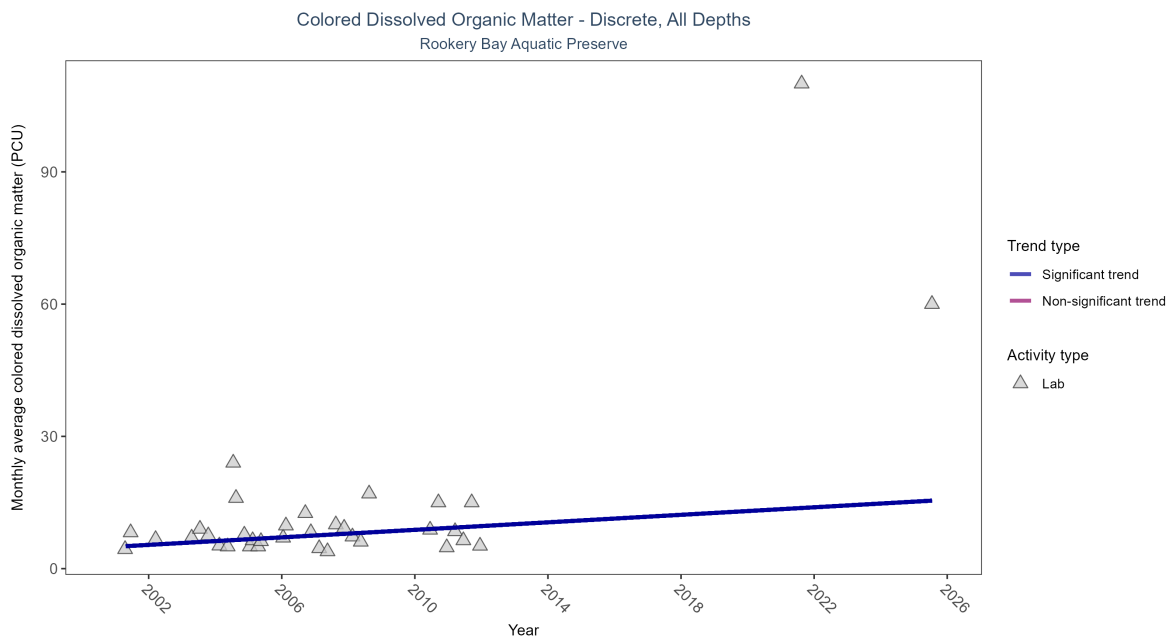


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Monthly average colored dissolved organic matter increased by 0.43 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	184	12	2001 - 2025	7	0.51961	4.96181	0.42619	0.03

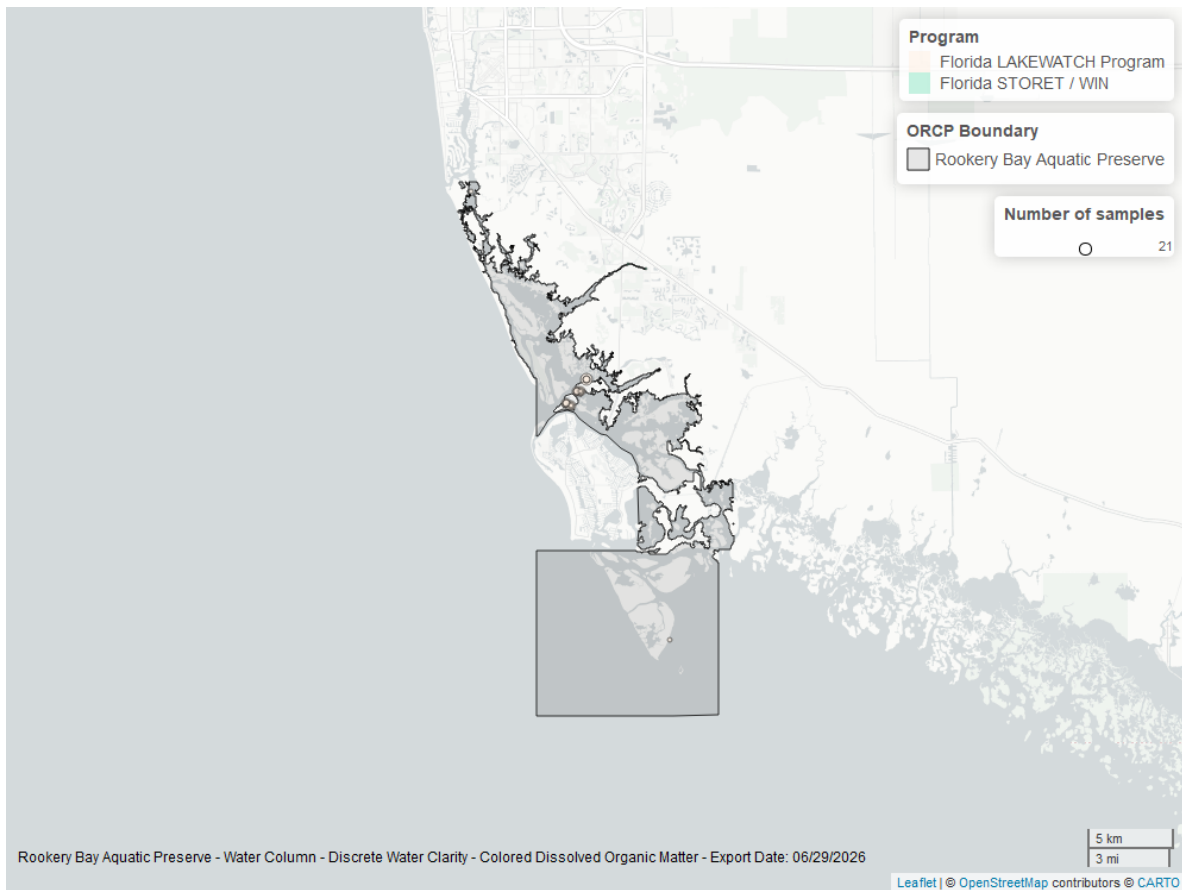


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.